

Addition 2: cross-architecture SHAP comparison

How to read this report. For each (target, feature set) cell and split type, the *headline* is the leaf chosen by a multi-metric composite - AUROC and AUPRC bucketed at a 0.02 noise tolerance, then calibration slope closest to 1 as the tie-break, with degenerate-calibration leaves excluded - and the *runner-up* is the best such leaf of a different architecture family whose AUROC CI overlaps the headline (the cross-architecture contrast). SHAP is computed on the calibrated positive-class probability (KernelSHAP for the XGBoost stack, the TabPFN-native explainer for TabPFN), so calibration quality is part of the selection. All metrics are hold-out test, with the data-split ratio, split type and hyperparameter-tuning configuration shown on every leaf line.

Split types. *Chronological* = forecasting-honest (train on the past) - the trustworthy section. *Stratified* = random shuffle, optimistically biased because history features carry adjacent-day signal across the train/test boundary (not deployable). *Patient hold-out* = generalisation to unseen patients. Attributions are comparable within a split type.

Key results

- **Best deployable (chronological) forecast:** migraine / full (xgboost, 70_30), AUROC 0.793 [0.701-0.873], AUPRC lift 4.8x.
- **Architecture and tuning:** 12 tabpfn, 9 xgboost across 21 headlines; library-default (NonHP) models win 20 of 21.
- **Stratified leakage:** excluded by split design; in 0 of 7 cells does the stratified AUROC CI separate above chronological, so no measured inflation is claimed.
- **Not above chance** (95% CI includes 0.5): headache/no_rolling/patient, migraine/park/patient.

Cross-family (XGBoost vs TabPFN) pair present: **yes**.

Best model per cell and split (headline)

The **chronological** column is the deployable, forecasting-honest best; **stratified** is not deployable - a random shuffle leaks adjacent-day signal through history features across the train/test boundary, so it is excluded by construction regardless of its measured AUROC; **patient** is generalisation to unseen patients. AUROC is hold-out test.

target / feature set	chronological (honest)	stratified (leaky)	patient (generalisation)
headache full_features	tabpfn 70_30 NonHP AUROC 0.652, calib +1.09	tabpfn 80_20 NonHP AUROC 0.700, calib +1.04	tabpfn 70_15_15 NonHP AUROC 0.646, calib +0.90
headache no_rolling_features	tabpfn 70_30 NonHP AUROC 0.582, calib +0.65	tabpfn 70_15_15 NonHP AUROC 0.573, calib +0.81	xgboost 80_20 NonHP AUROC 0.522, calib +1.00
headache spano_features	xgboost 70_30 NonHP AUROC 0.638, calib +1.49	xgboost 70_15_15 NonHP AUROC 0.664, calib +1.38	xgboost 70_30 NonHP AUROC 0.631, calib +1.29

migraine full_features	xgboost 70_30 <i>single_AUROC</i> AUROC 0.793, calib +1.42	tabpfn 70_15_15 <i>NonHP</i> AUROC 0.736, calib +1.01	tabpfn 70_15_15 <i>NonHP</i> AUROC 0.716, calib +0.71
migraine no_rolling_features	tabpfn 70_15_15 <i>NonHP</i> AUROC 0.749, calib +0.71	tabpfn 70_30 <i>NonHP</i> AUROC 0.614, calib +0.57	tabpfn 70_30 <i>NonHP</i> AUROC 0.599, calib +0.42
migraine park_features	tabpfn 70_30 <i>NonHP</i> AUROC 0.646, calib +1.05	tabpfn 80_20 <i>NonHP</i> AUROC 0.618, calib +0.59	xgboost 70_30 <i>NonHP</i> AUROC 0.541, calib +0.67
migraine spano_features	xgboost 70_30 <i>NonHP</i> AUROC 0.770, calib +1.90	xgboost 70_15_15 <i>NonHP</i> AUROC 0.716, calib +0.94	xgboost 80_20 <i>NonHP</i> AUROC 0.651, calib +4.96

What wins the headline, overall

Headline architecture family (count of cells won per split):

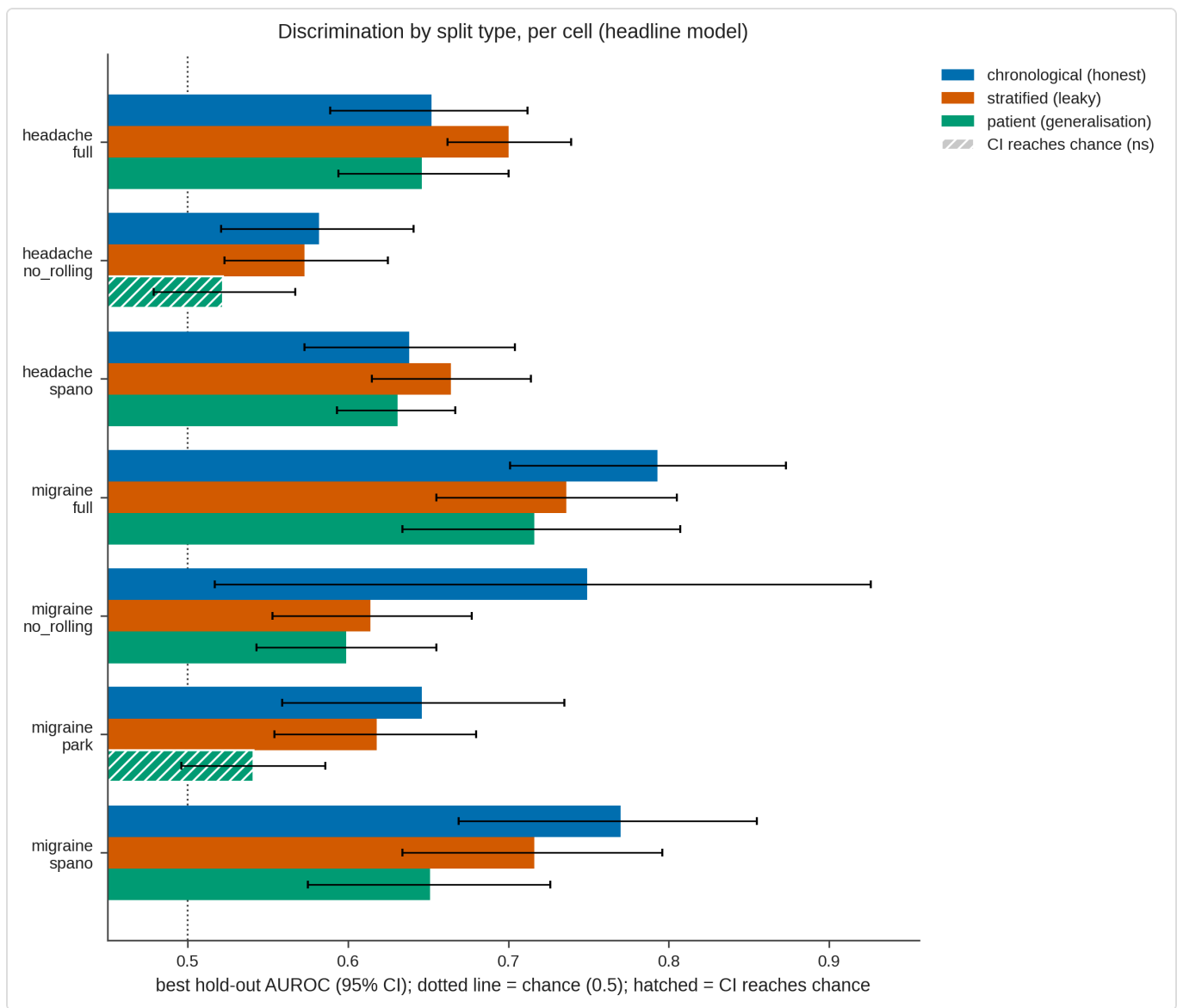
	chronological	stratified	patient	total
tabpfn	4	5	3	12
xgboost	3	2	4	9

Headline hyperparameter-tuning strategy:

	chronological	stratified	patient	total
NonHP (library defaults)	6	7	7	20
single_AUROC/HP020	1	0	0	1

Discrimination across split types

Best hold-out AUROC per cell, one bar per split. The **chronological** bar is the deployable forecast; **stratified** is the leakage contrast - a random shuffle places adjacent days, which share history / rolling feature values, on both sides of the train/test boundary; **patient** is generalisation to unseen patients. The leakage is a property of the split design, so stratified is excluded by construction; the magnitude of any empirical inflation is reported separately below, since the wide CIs at this sample size do not by themselves establish it.

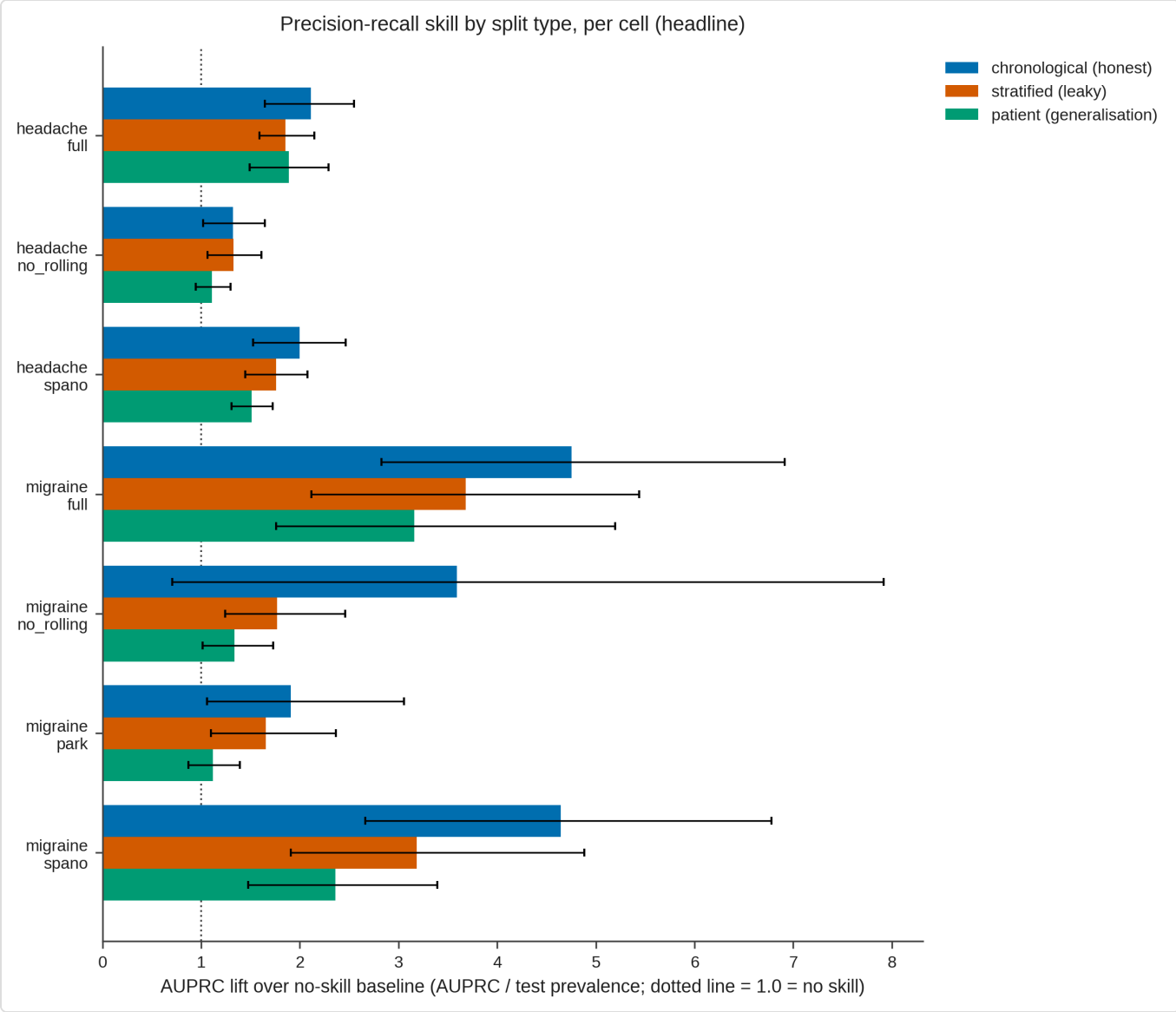


In no cell does the stratified CI separate from the chronological one, so the data do not establish a measurable inflation at this sample size; the stratified split is excluded for its leakage mechanism, not for an observed inflation.

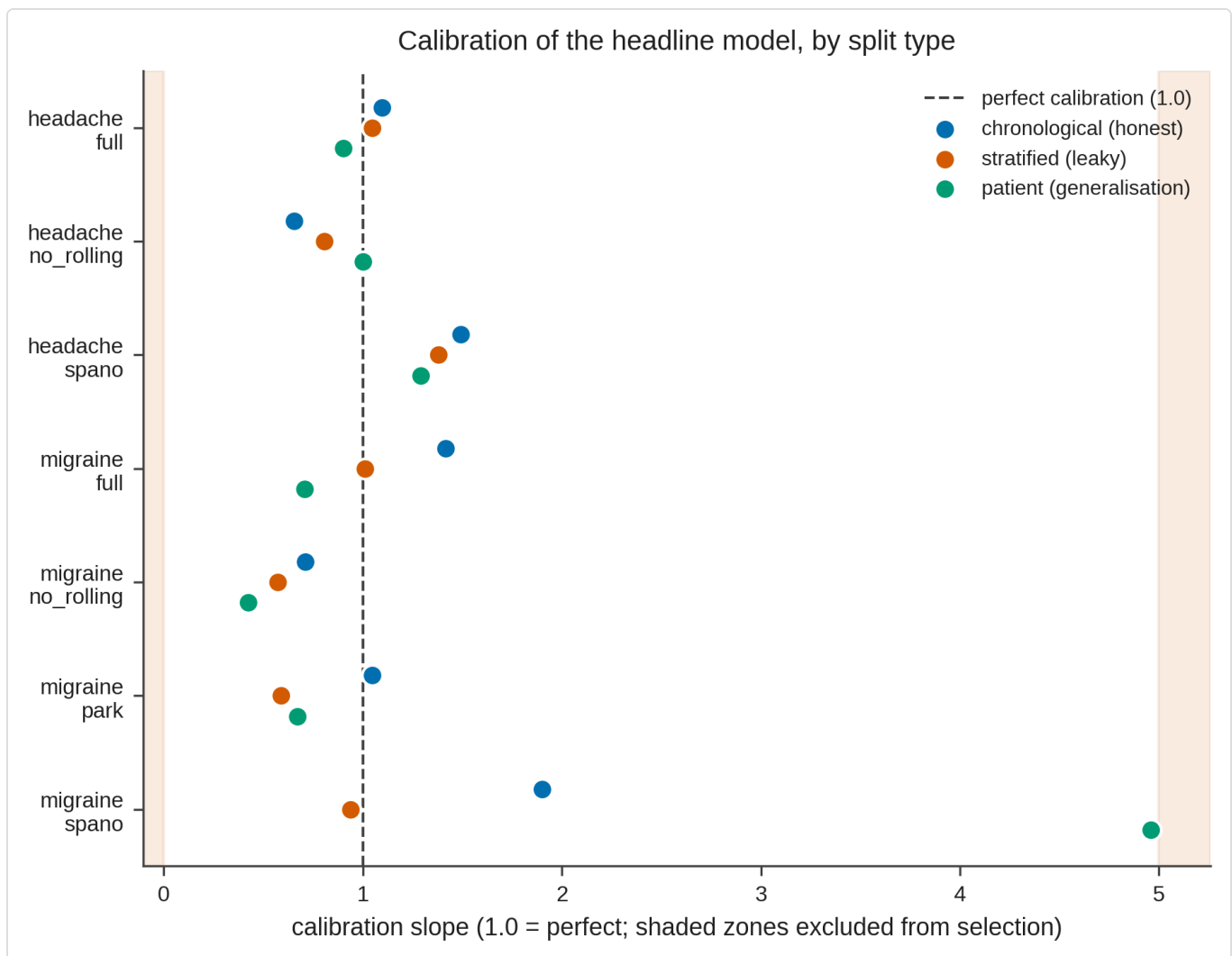
cell	chronological AUROC	stratified AUROC	Δ	CI verdict
headache / full	0.652 [0.589-0.712]	0.700 [0.662-0.739]	+0.048	not distinguishable (CIs overlap)
headache / no_rolling	0.582 [0.521-0.641]	0.573 [0.523-0.625]	-0.009	not distinguishable (CIs overlap)
headache / spano	0.638 [0.573-0.704]	0.664 [0.615-0.714]	+0.026	not distinguishable (CIs overlap)
migraine / full	0.793 [0.701-0.873]	0.736 [0.655-0.805]	-0.057	not distinguishable (CIs overlap)
migraine / no_rolling	0.749 [0.517-0.926]	0.614 [0.553-0.677]	-0.135	not distinguishable (CIs overlap)
migraine / park	0.646 [0.559-0.735]	0.618 [0.554-0.680]	-0.028	not distinguishable (CIs overlap)

migraine / spano	0.770 [0.669-0.855]	0.716 [0.634-0.796]	-0.054	not distinguishable (CIs overlap)
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Under the heavy class imbalance (the migraine positive rate is ~5-7 %), AUROC can look respectable while precision-recall stays near the base rate. AUPRC lift = AUPRC divided by the test-set positive prevalence puts both targets on a common scale; **1.0 is no skill** (no better than predicting the base rate). Each bar uses its own leaf's test prevalence as the baseline.



Calibration is the second axis of forecast quality: discrimination ranks days, calibration scales the probabilities. The selection prefers a slope near 1.0 and excludes the shaded (inverted or mis-scaled) zones, so a cell sitting far from 1.0 discriminates without yielding trustworthy probabilities.



Chronological (forecasting-honest)

headache / full_features

headline: **tabpfn/version_2-6** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.652, AUPRC 0.411, calib +1.09

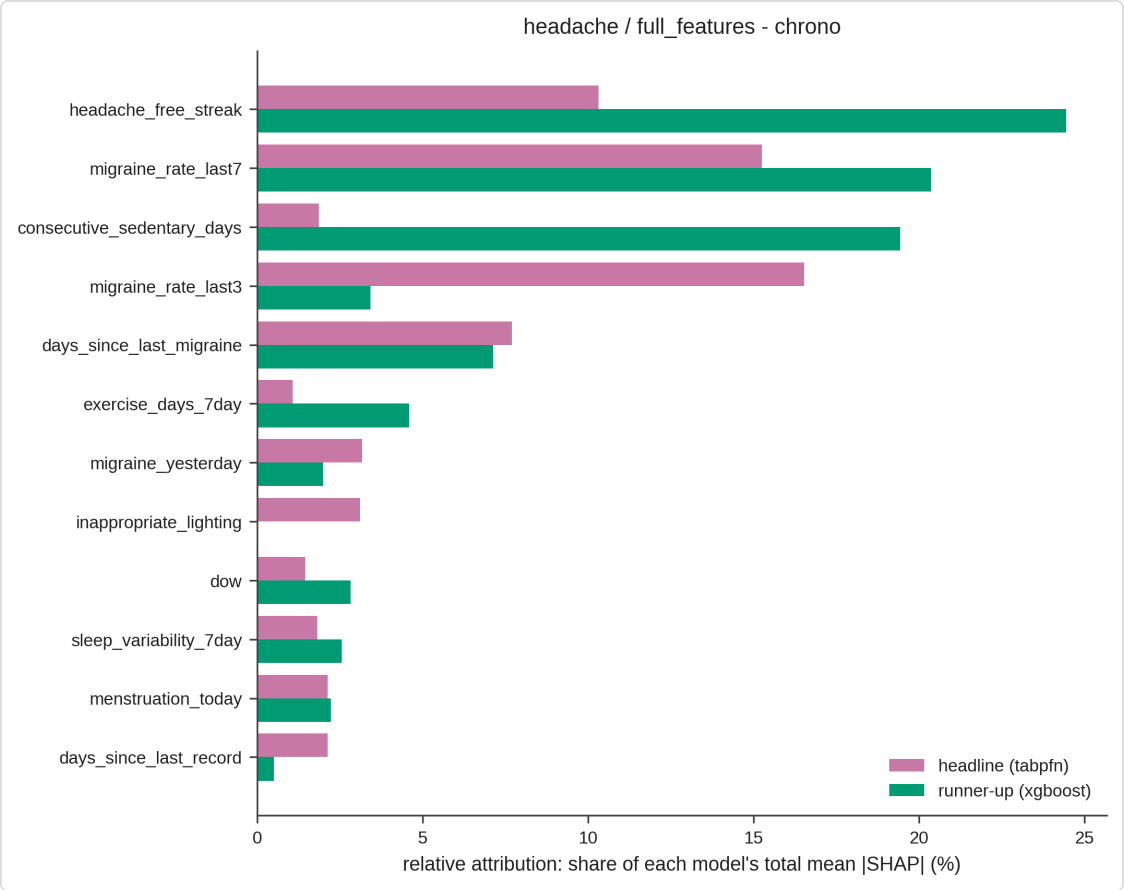
runner-up: **stacked_2xgb_meta_lr** [single_AUROC/HP020] | split 70_30/chrono | hold-out AUROC 0.657, AUPRC 0.404, calib +1.41

Spearman rank correlation over all 52 shared features: **+0.620** - moderate cross-architecture agreement on feature ordering.

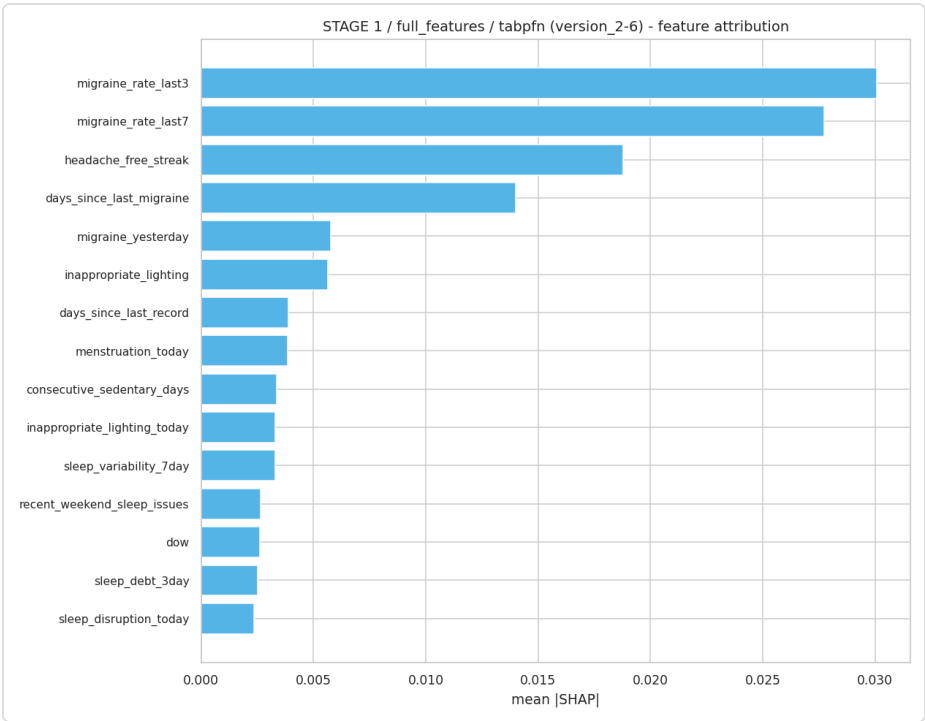
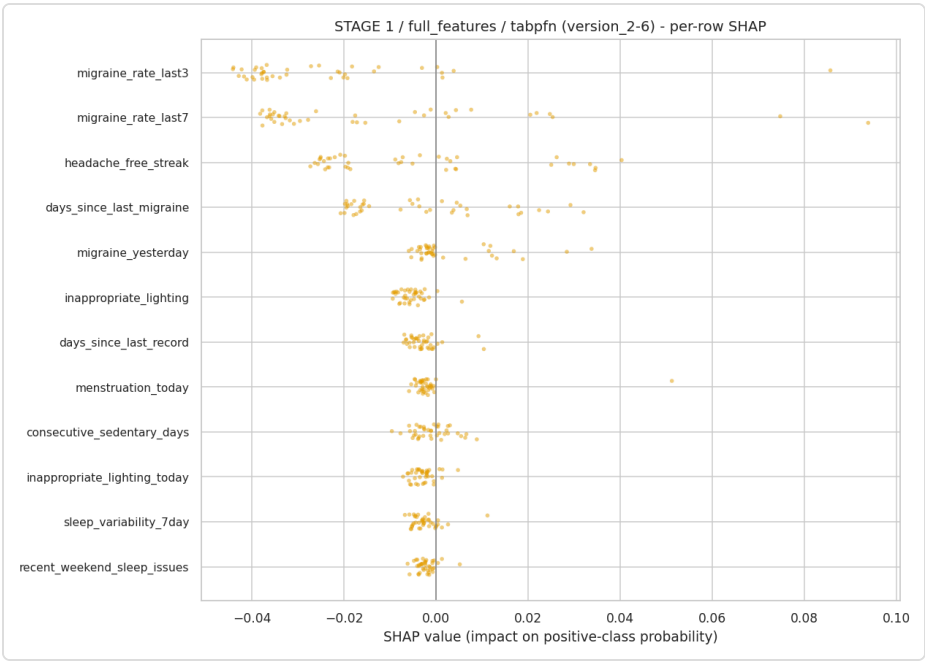
rank	headline (tabpfn)	runner-up (xgboost)
1	migraine_rate_last3	headache_free_streak
2	migraine_rate_last7	migraine_rate_last7
3	headache_free_streak	consecutive_sedentary_days
4	days_since_last_migraine	days_since_last_migraine
5	migraine_yesterday	exercise_days_7day
6	inappropriate_lighting	migraine_rate_last3
7	days_since_last_record	dow

8	menstruation_today	sleep_variability_7day
9	consecutive_sedentary_days	menstruation_today
10	inappropriate_lighting_today	lack_of_sleep_today

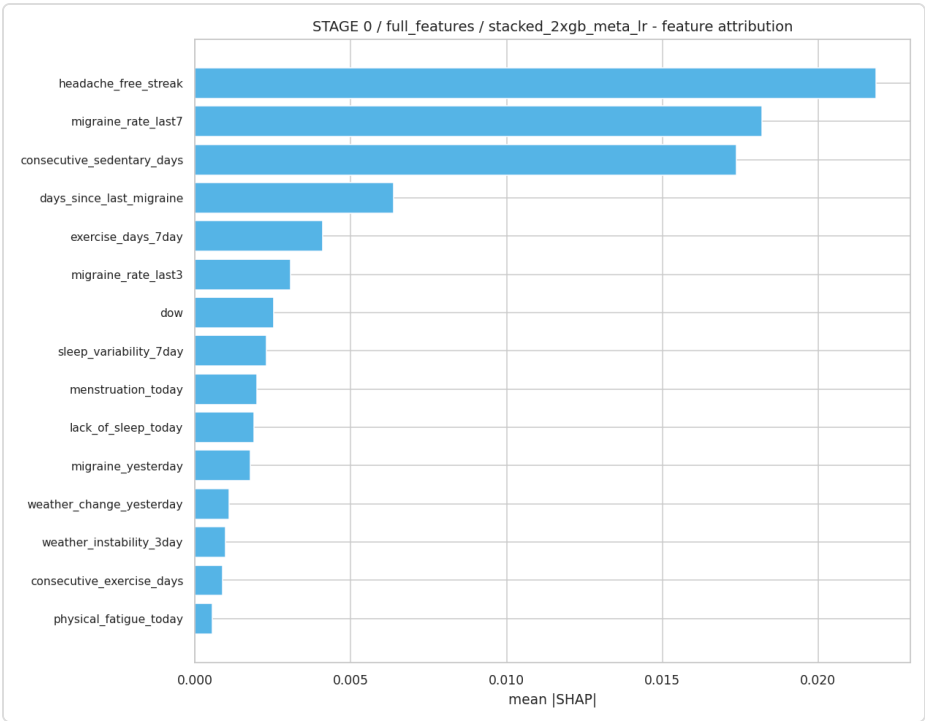
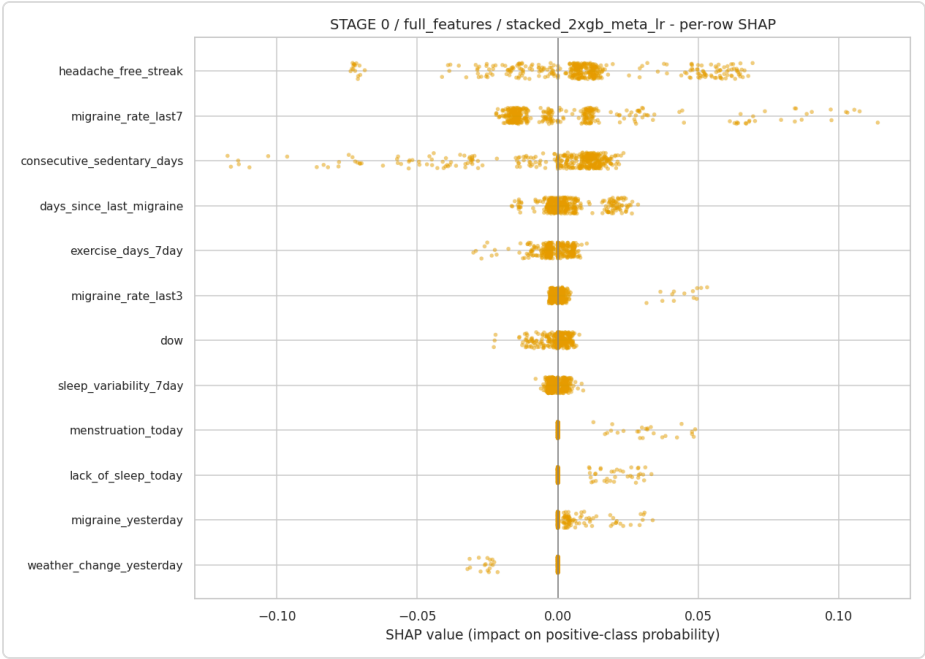
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_30/chrono)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/chrono)



headache / no_rolling_features

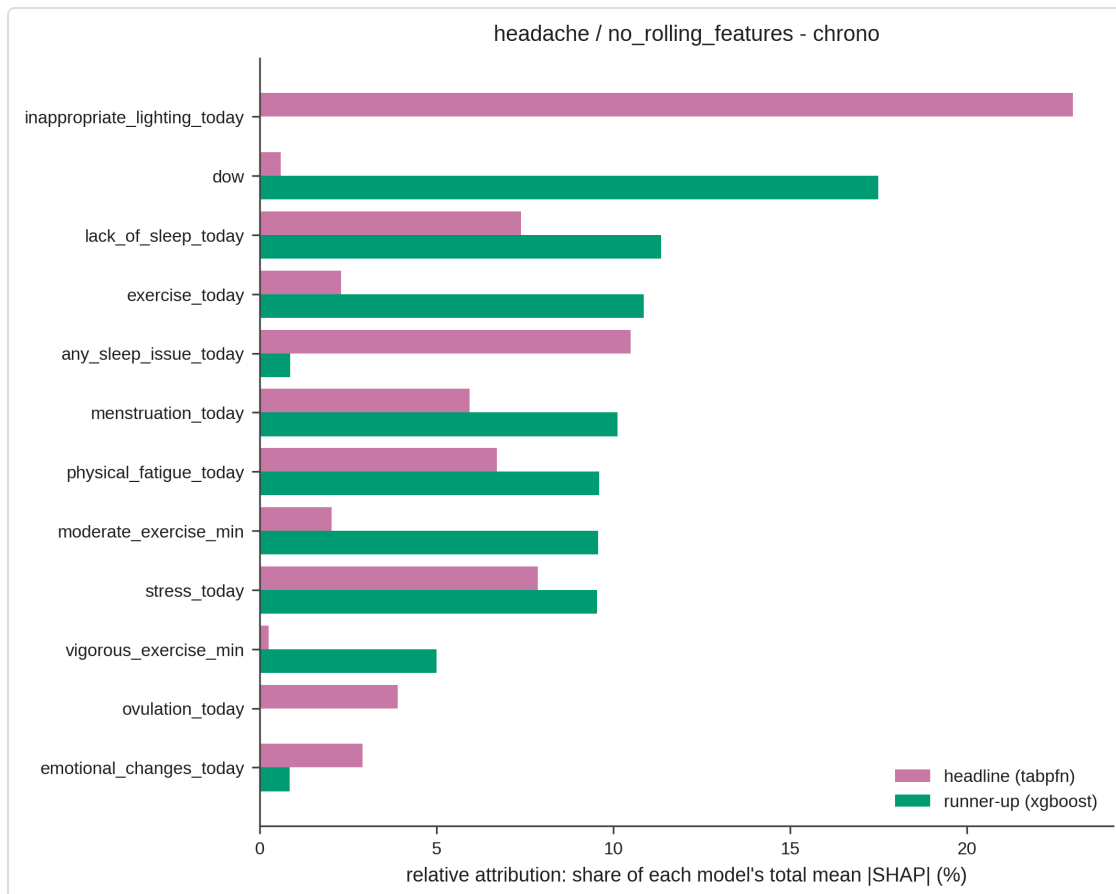
headline: **tabpfm/version_3-binary** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.582, AUPRC 0.258, calib +0.65
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.553, AUPRC 0.246, calib +0.80

Spearman rank correlation over all 26 shared features: **-0.011** - weak cross-architecture disagreement on feature ordering.

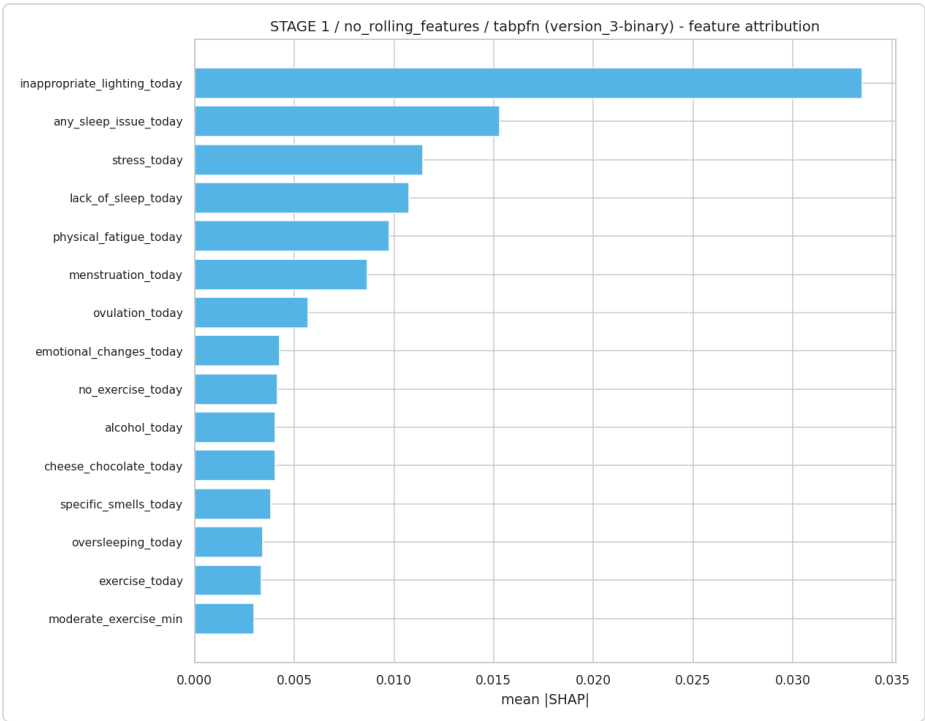
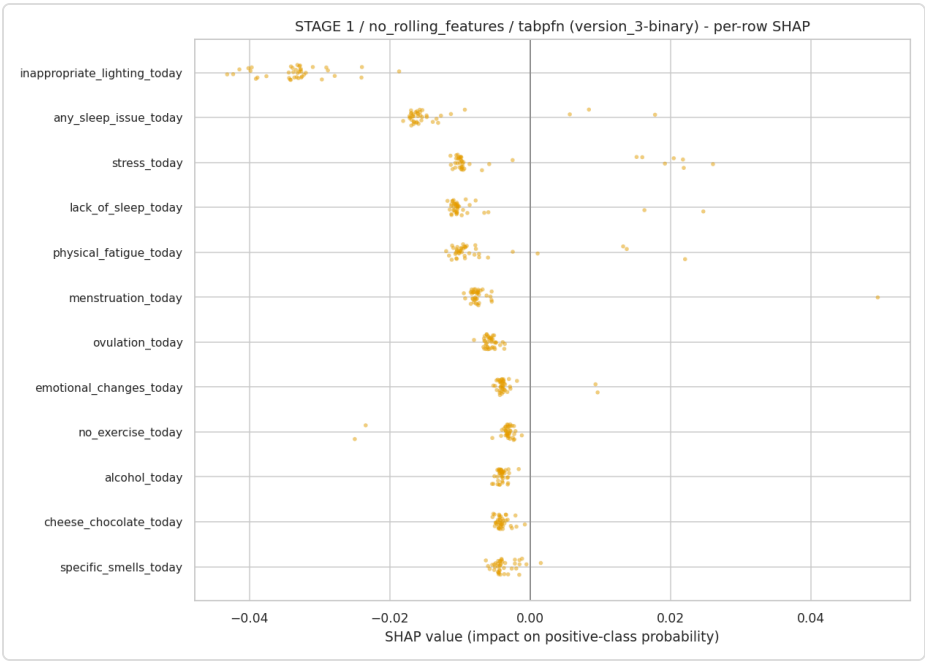
rank	headline (tabpfm)	runner-up (xgboost)
1	inappropriate_lighting_today	dow

2	any_sleep_issue_today	lack_of_sleep_today
3	stress_today	exercise_today
4	lack_of_sleep_today	menstruation_today
5	physical_fatigue_today	physical_fatigue_today
6	menstruation_today	moderate_exercise_min
7	ovulation_today	stress_today
8	emotional_changes_today	vigorous_exercise_min
9	no_exercise_today	excessive_caffeine_today
10	alcohol_today	no_exercise_today

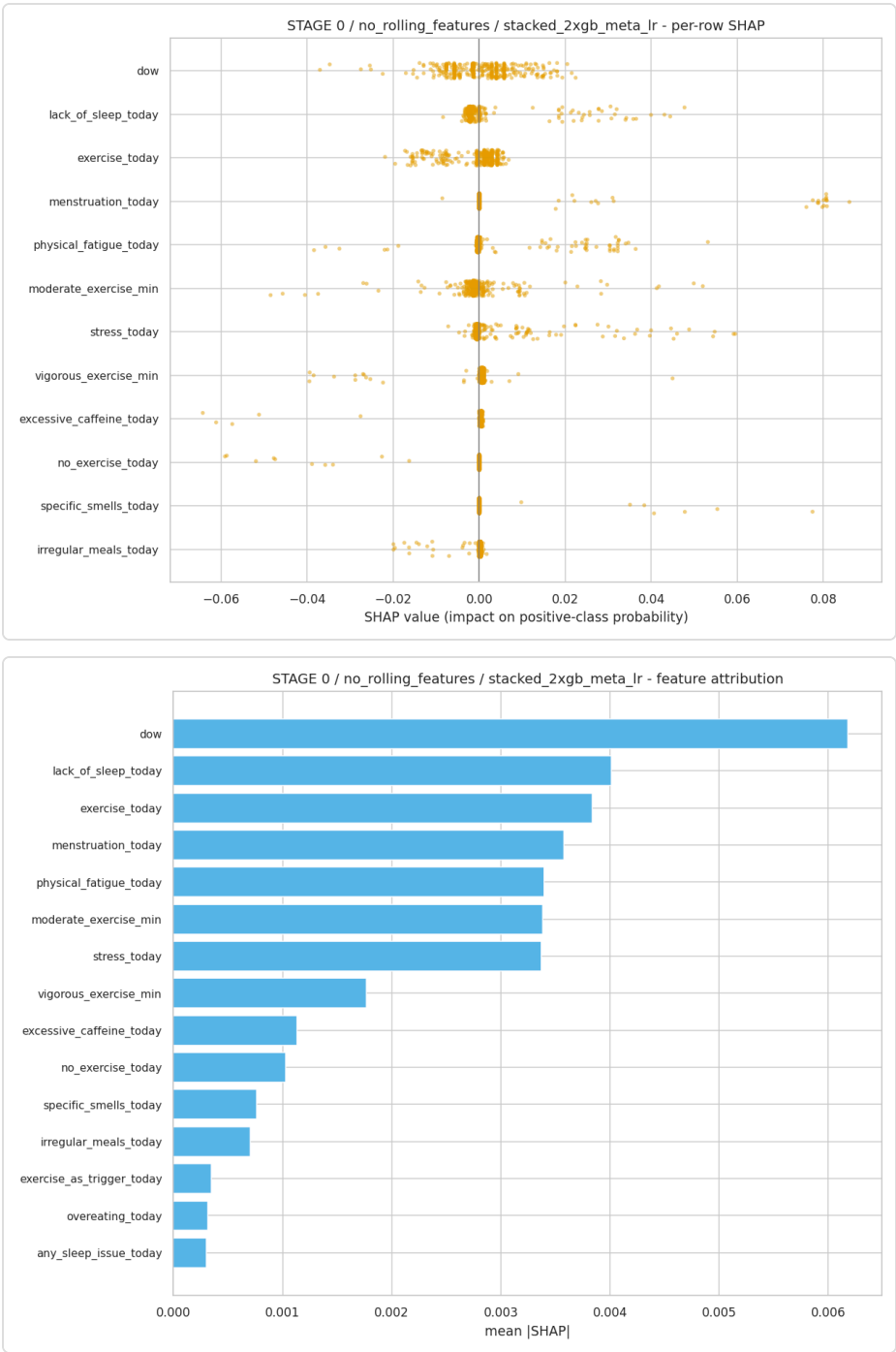
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_30/chrono)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/chrono)



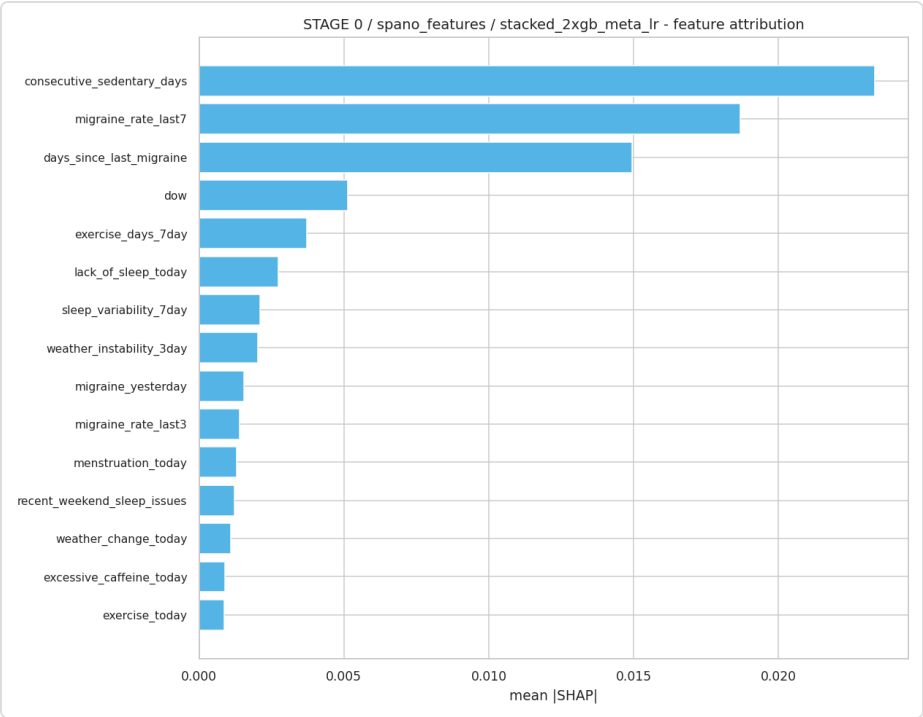
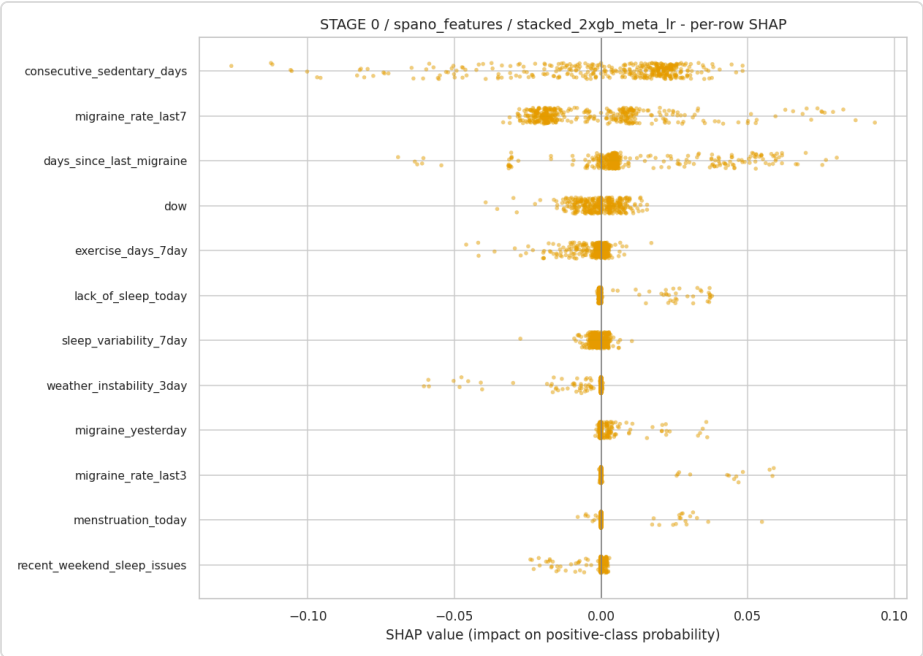
headache / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.638, AUPRC 0.389, calib +1.49. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	consecutive_sedentary_days	0.0233
2	migraine_rate_last7	0.0187
3	days_since_last_migraine	0.0149
4	dow	0.0051

5	exercise_days_7day	0.0037
6	lack_of_sleep_today	0.0027
7	sleep_variability_7day	0.0021
8	weather_instability_3day	0.0020
9	migraine_yesterday	0.0015
10	migraine_rate_last3	0.0014

leaf: xgboost stacked_2xgb_meta_lr (70_30/chrono)



migraine / full_features

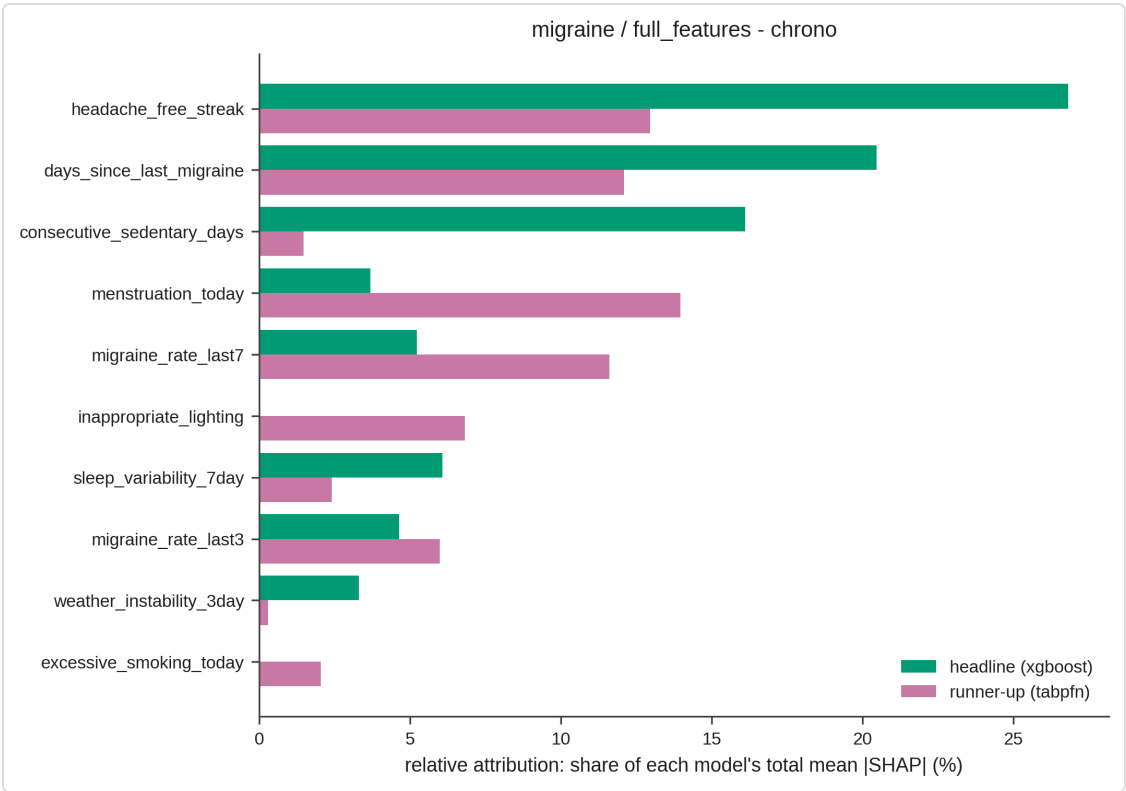
headline: **stacked_2xgb_meta_lr** [single_AUROC/HP020] | split 70_30/chrono | hold-out AUROC 0.793, AUPRC 0.314, calib +1.42

runner-up: **tabpfn/version_2-5-finetuned** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.761, AUPRC 0.320, calib +0.96

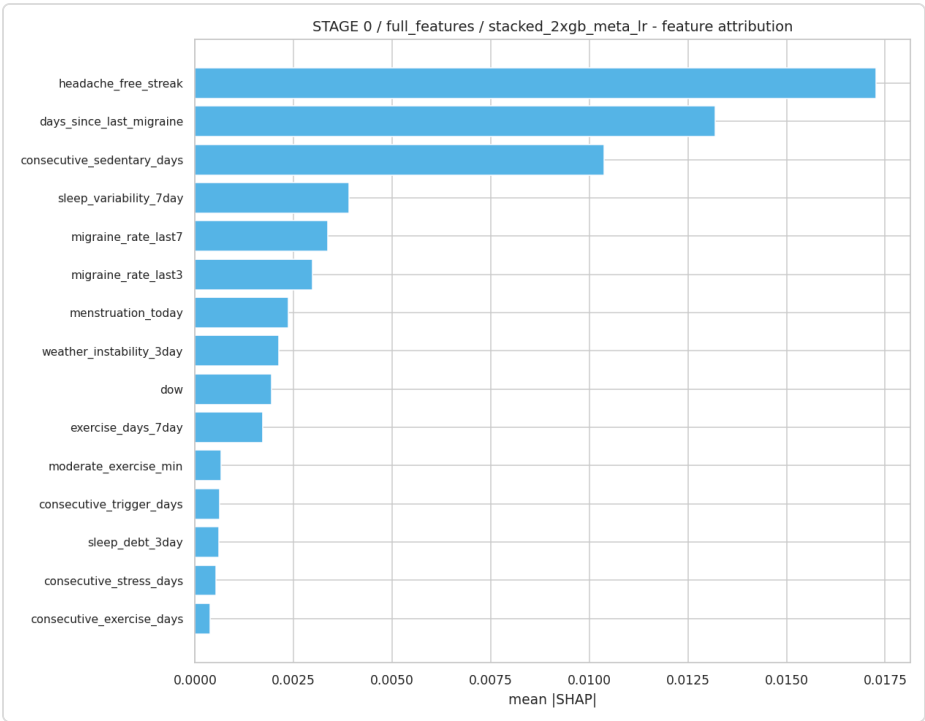
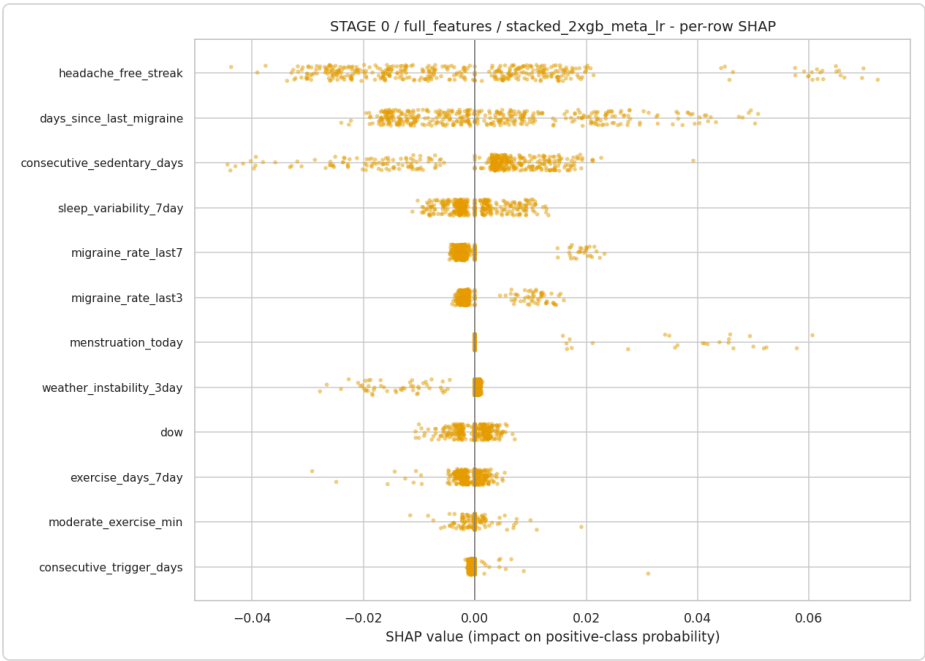
Spearman rank correlation over all 52 shared features: **+0.171** - weak cross-architecture agreement on feature ordering.

rank	headline (xgboost)	runner-up (tabpfn)
1	headache_free_streak	menstruation_today
2	days_since_last_migraine	headache_free_streak
3	consecutive_sedentary_days	days_since_last_migraine
4	sleep_variability_7day	migraine_rate_last7
5	migraine_rate_last7	inappropriate_lighting
6	migraine_rate_last3	migraine_rate_last3
7	menstruation_today	sleep_variability_7day
8	weather_instability_3day	excessive_smoking_today
9	dow	inappropriate_lighting_today
10	exercise_days_7day	exercise_days_7day

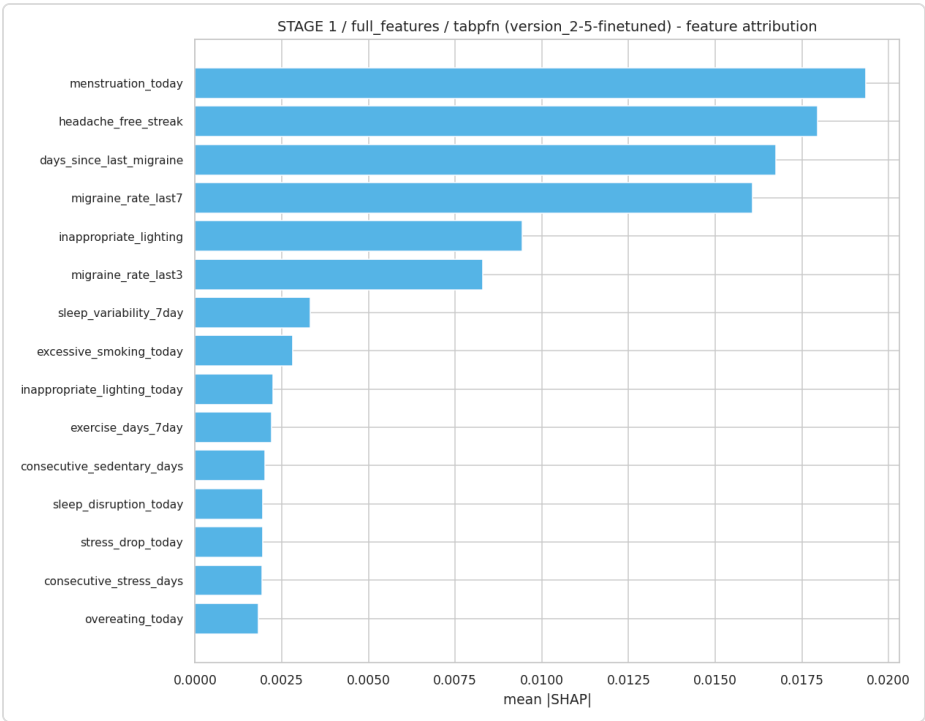
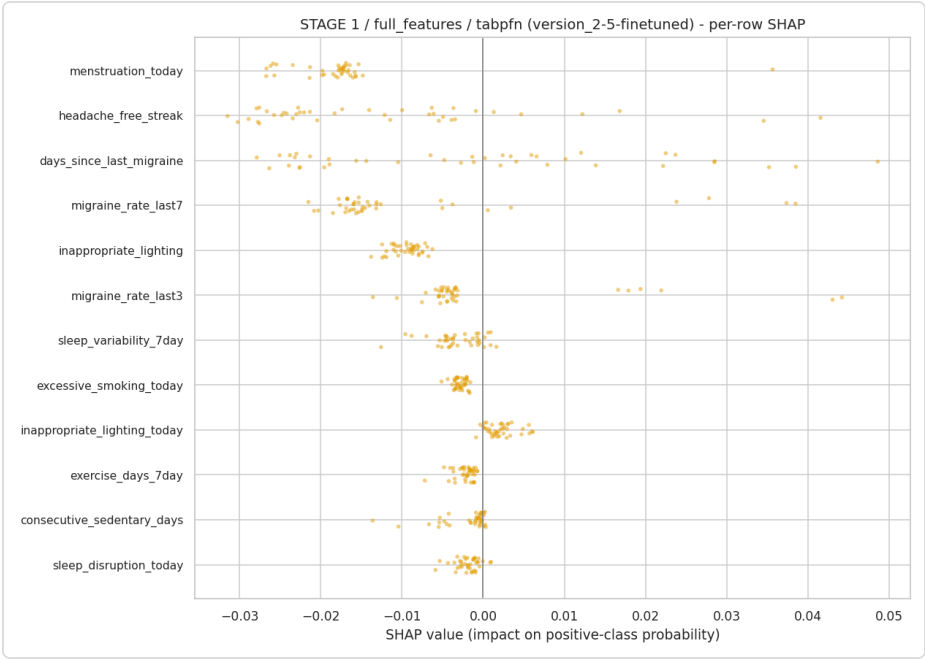
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: xgboost stacked_2xgb_meta_lr (70_30/chrono)



runner-up: tabpfn tabpfn (70_30/chrono)



migraine / no_rolling_features

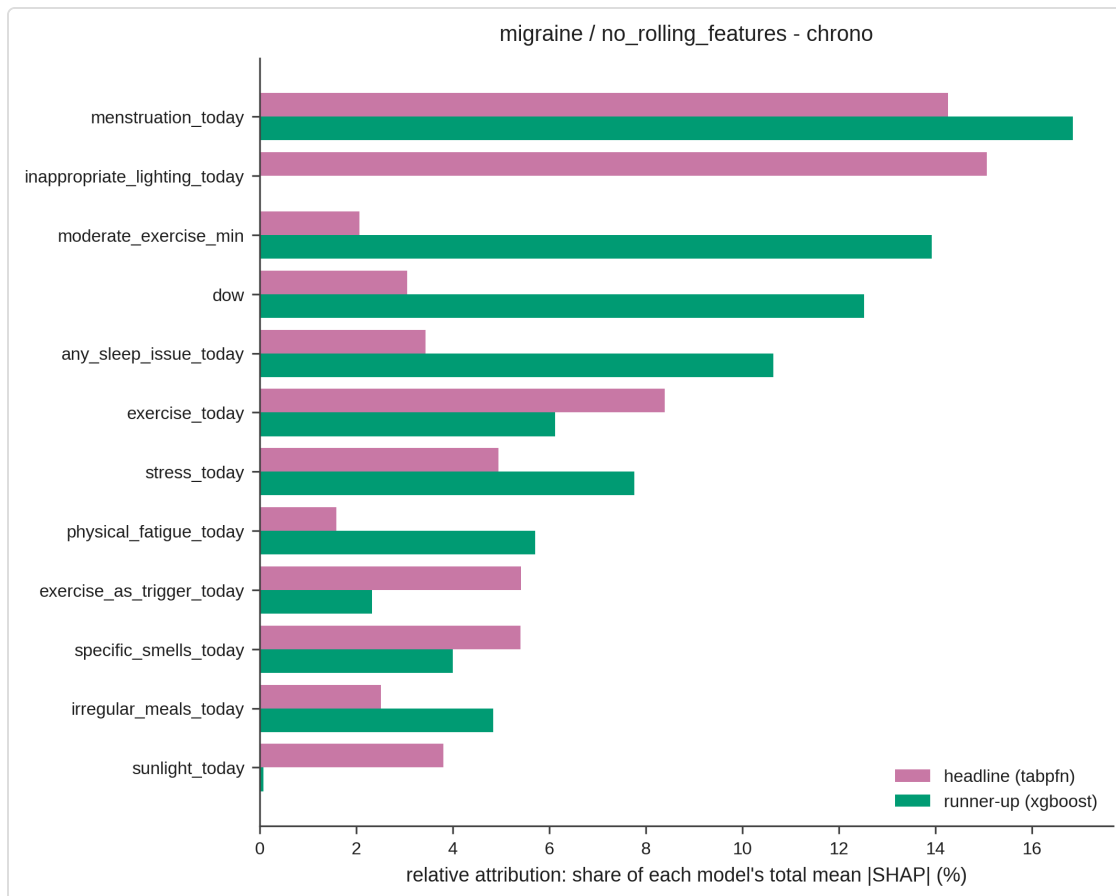
headline: **tabpfn/version_2-5-real** [NonHP (library defaults)] | split 70_15_15/chrono | hold-out AUROC 0.749, AUPRC 0.132, calib +0.71
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.619, AUPRC 0.155, calib +1.63

Spearman rank correlation over all 26 shared features: **+0.216** - weak cross-architecture agreement on feature ordering.

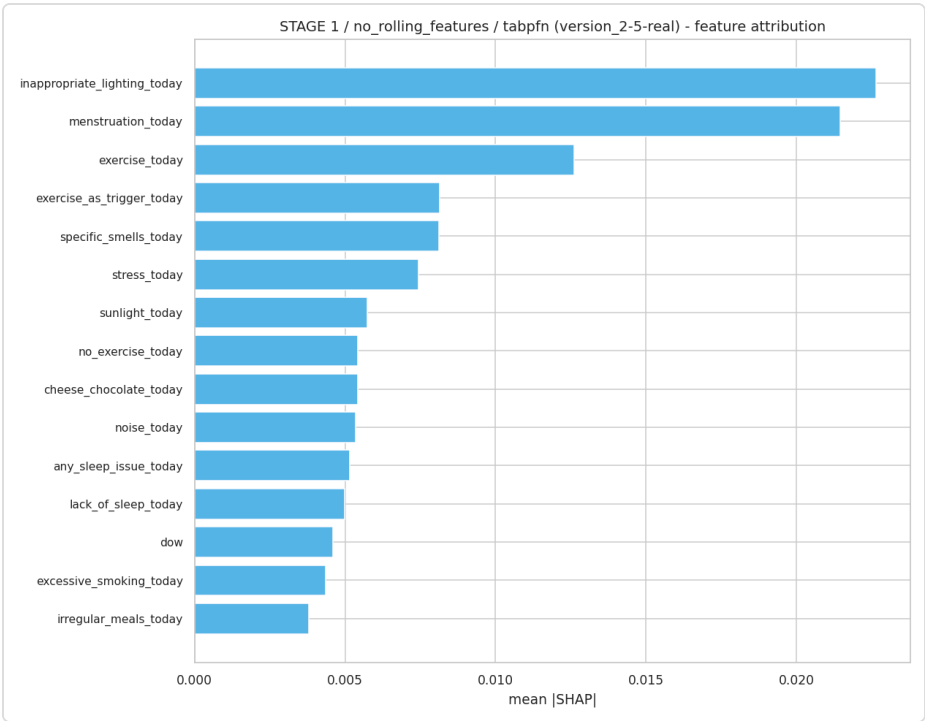
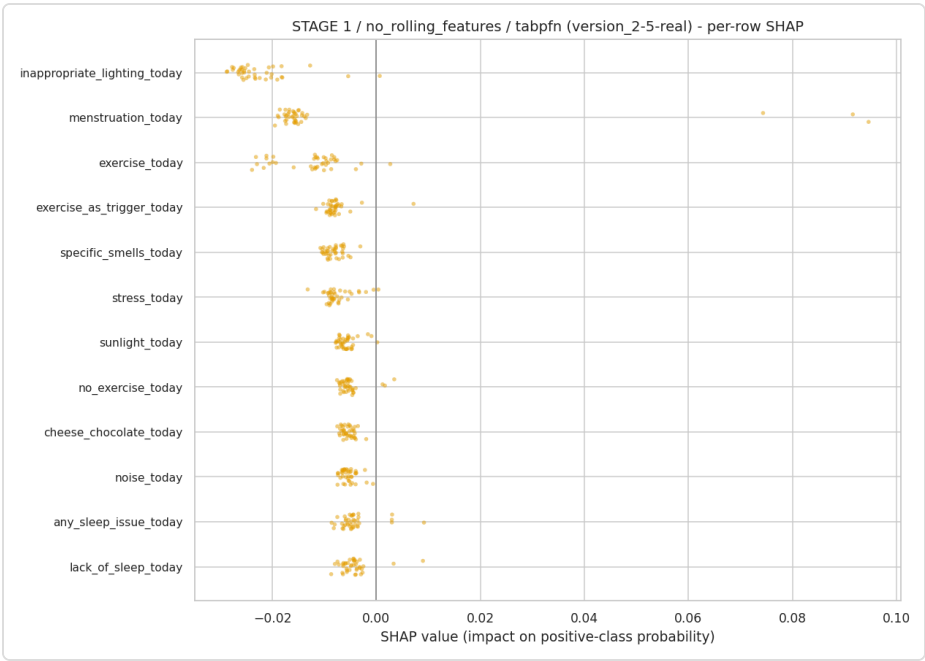
rank	headline (tabpfn)	runner-up (xgboost)
1	inappropriate_lighting_today	menstruation_today

2	menstruation_today	moderate_exercise_min
3	exercise_today	dow
4	exercise_as_trigger_today	any_sleep_issue_today
5	specific_smells_today	stress_today
6	stress_today	exercise_today
7	sunlight_today	physical_fatigue_today
8	no_exercise_today	irregular_meals_today
9	cheese_chocolate_today	lack_of_sleep_today
10	noise_today	specific_smells_today

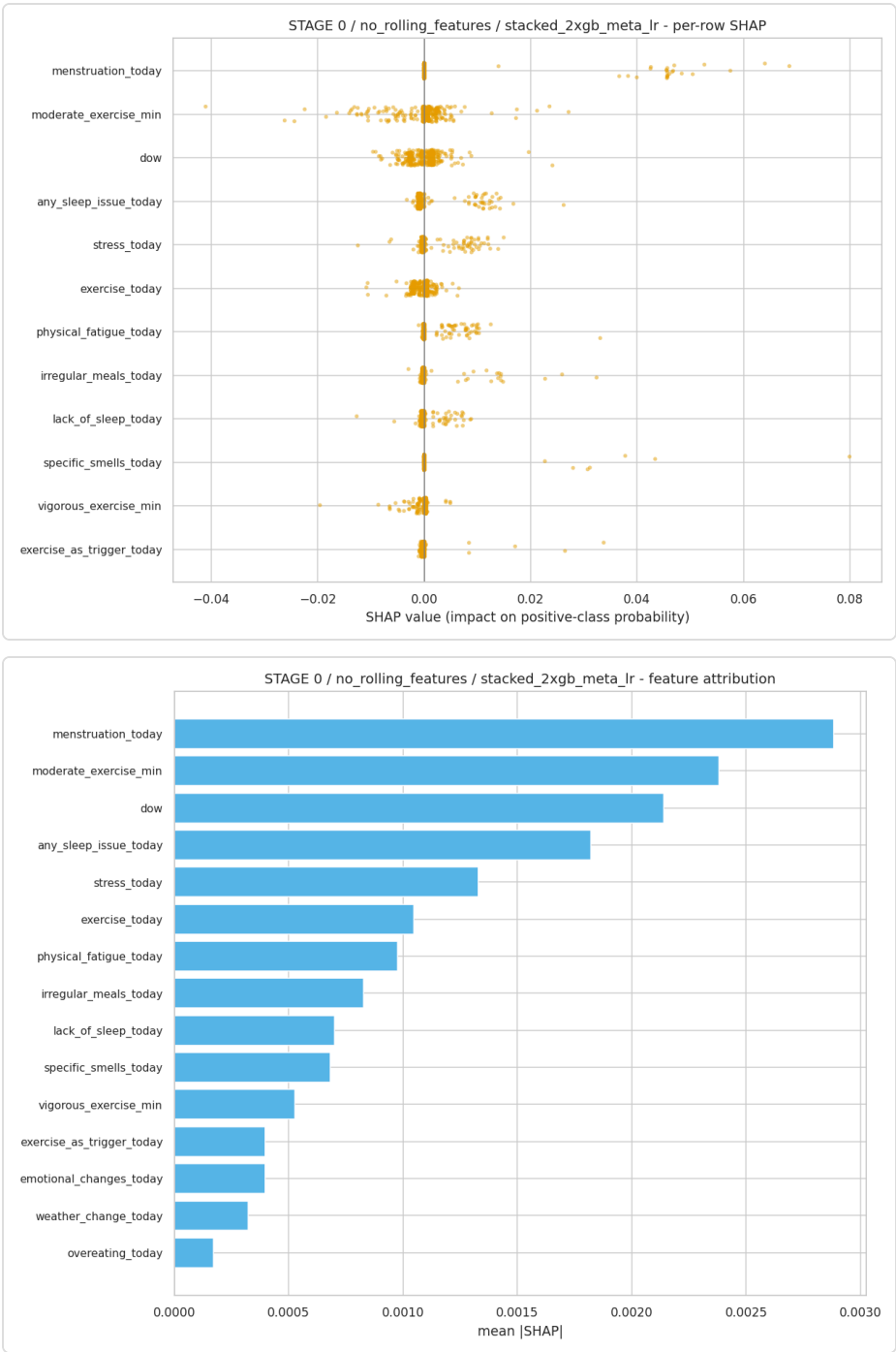
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_15_15/chrono)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/chrono)



migraine / park_features

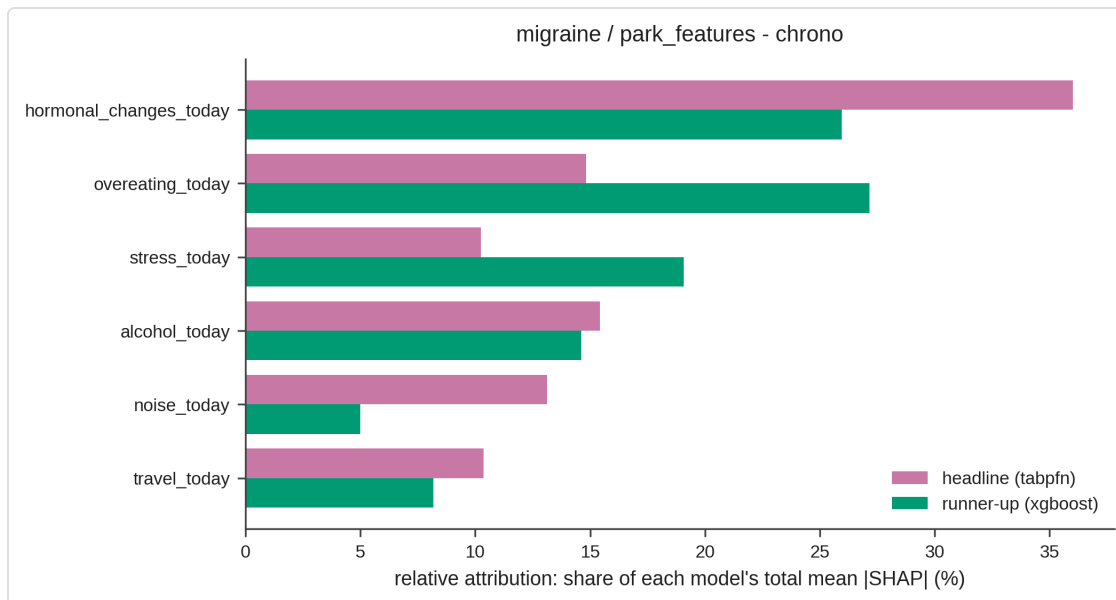
headline: **tabpfnn/version_3-binary** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.646, AUPRC 0.126, calib +1.05
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.625, AUPRC 0.115, calib +0.82

Spearman rank correlation over all 6 shared features: **+0.371** - weak cross-architecture agreement on feature ordering.

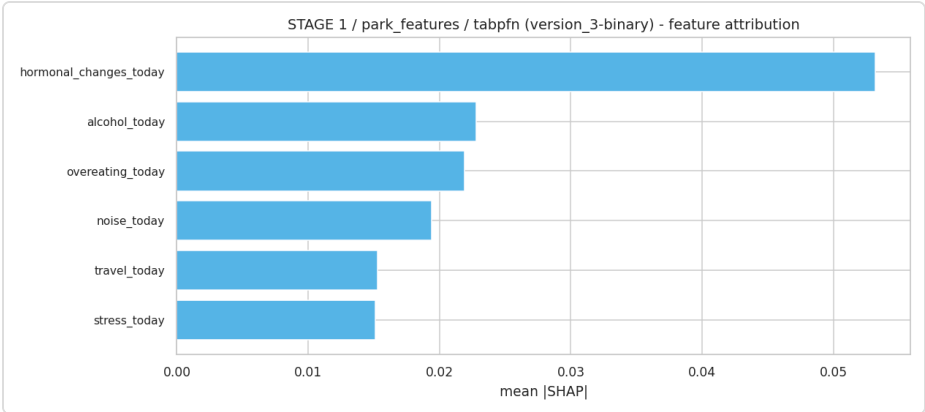
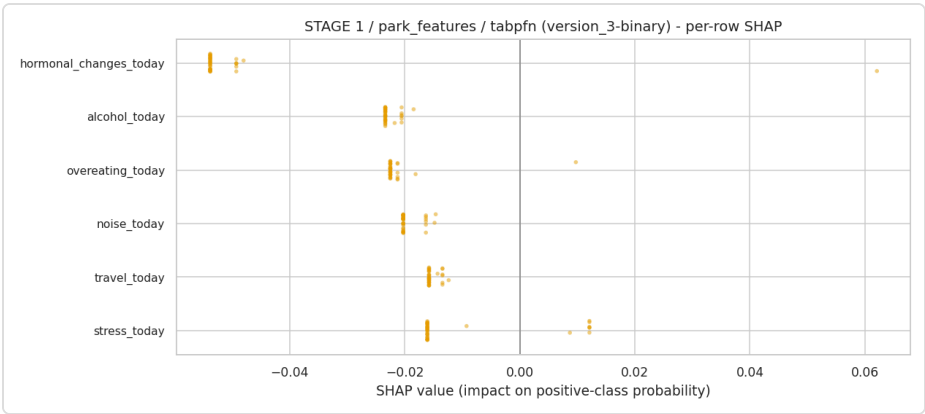
rank	headline (tabpfnn)	runner-up (xgboost)
1	hormonal_changes_today	overeating_today

2	alcohol_today	hormonal_changes_today
3	overeating_today	stress_today
4	noise_today	alcohol_today
5	travel_today	travel_today
6	stress_today	noise_today
7		
8		
9		
10		

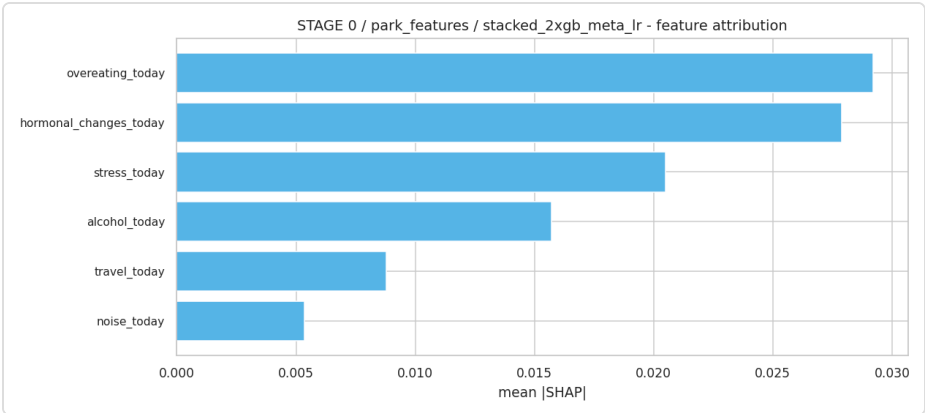
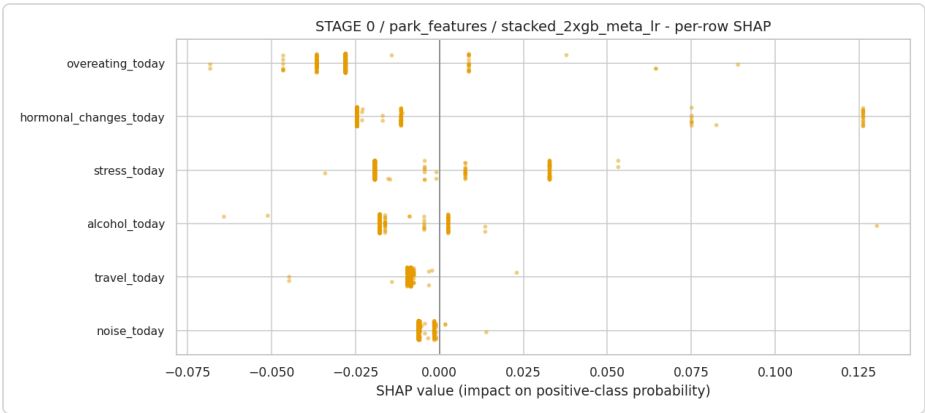
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_30/chrono)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/chrono)

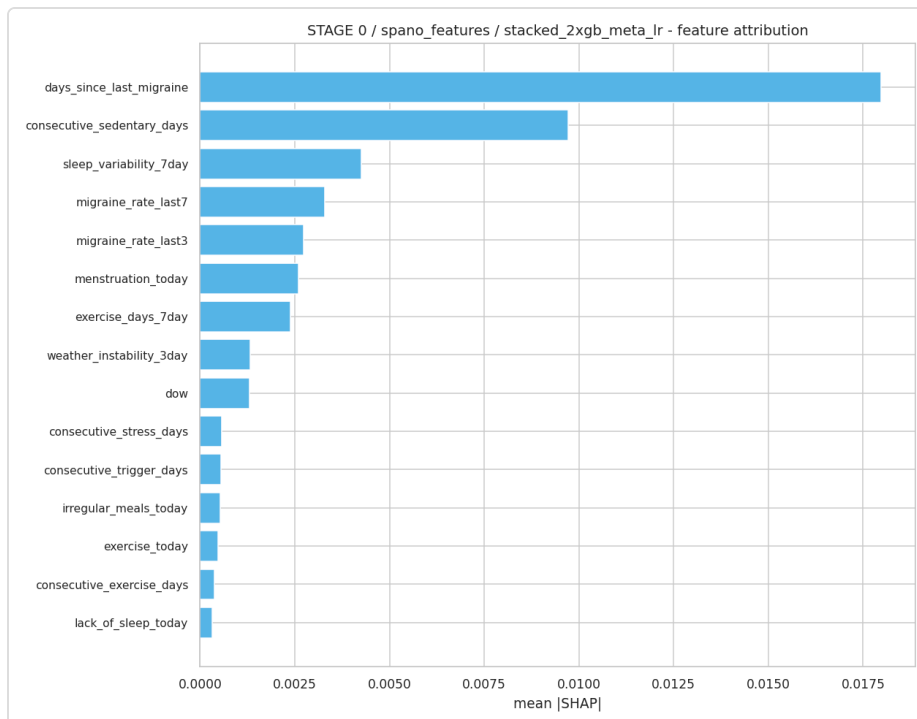
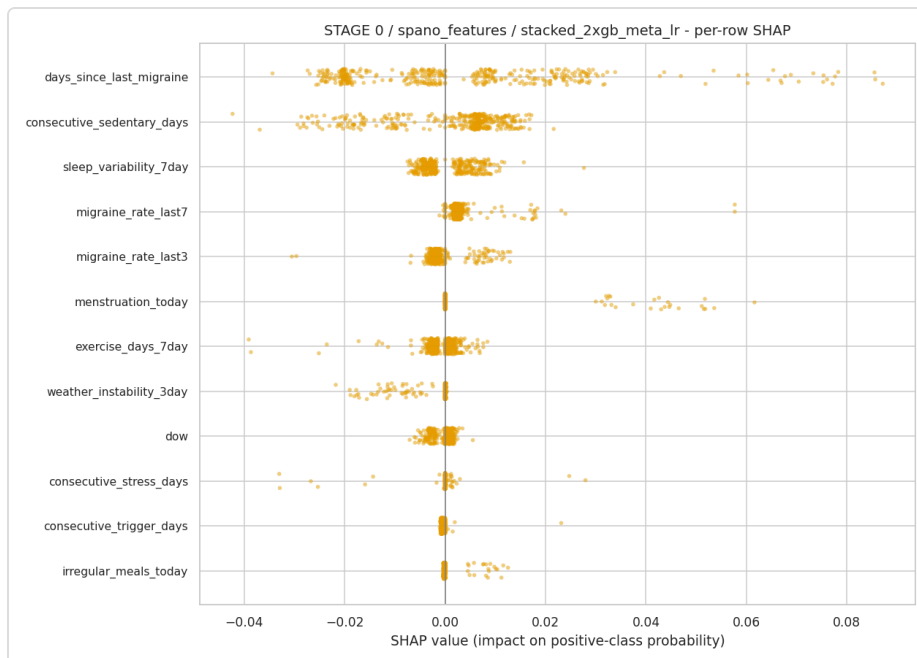


migraine / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/chrono | hold-out AUROC 0.770, AUPRC 0.307, calib +1.90. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	days_since_last_migraine	0.0180
2	consecutive_sedentary_days	0.0097
3	sleep_variability_7day	0.0043
4	migraine_rate_last7	0.0033
5	migraine_rate_last3	0.0027
6	menstruation_today	0.0026
7	exercise_days_7day	0.0024
8	weather_instability_3day	0.0013
9	dow	0.0013
10	consecutive_stress_days	0.0006

leaf: xgboost stacked_2xgb_meta_lr (70_30/chrono)



Stratified (leakage contrast - not deployable by construction)

headache / full_features

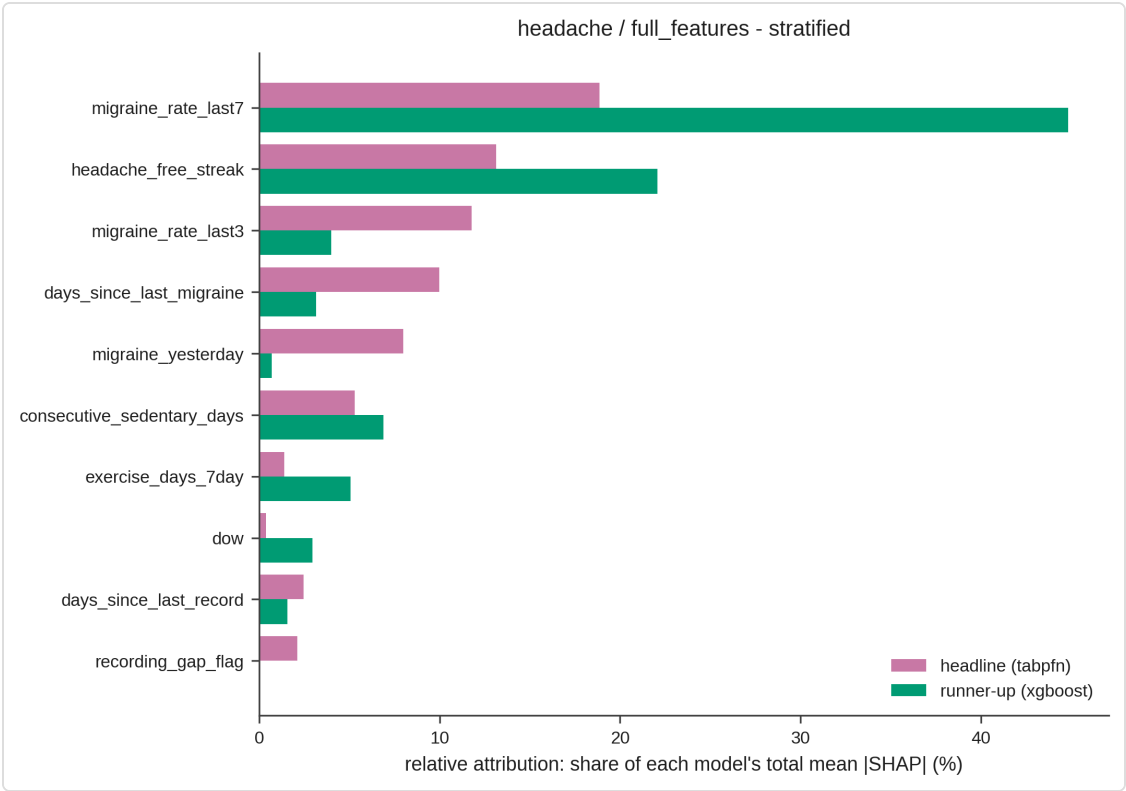
headline: **tabpfn/version_3-default** [NonHP (library defaults)] | split 80_20/stratified | hold-out AUROC 0.700, AUPRC 0.435, calib +1.04

runner-up: **stacked_2xgb_meta_lr** [single_AUROC/HP200] | split 70_15_15/stratified | hold-out AUROC 0.679, AUPRC 0.446, calib +1.28

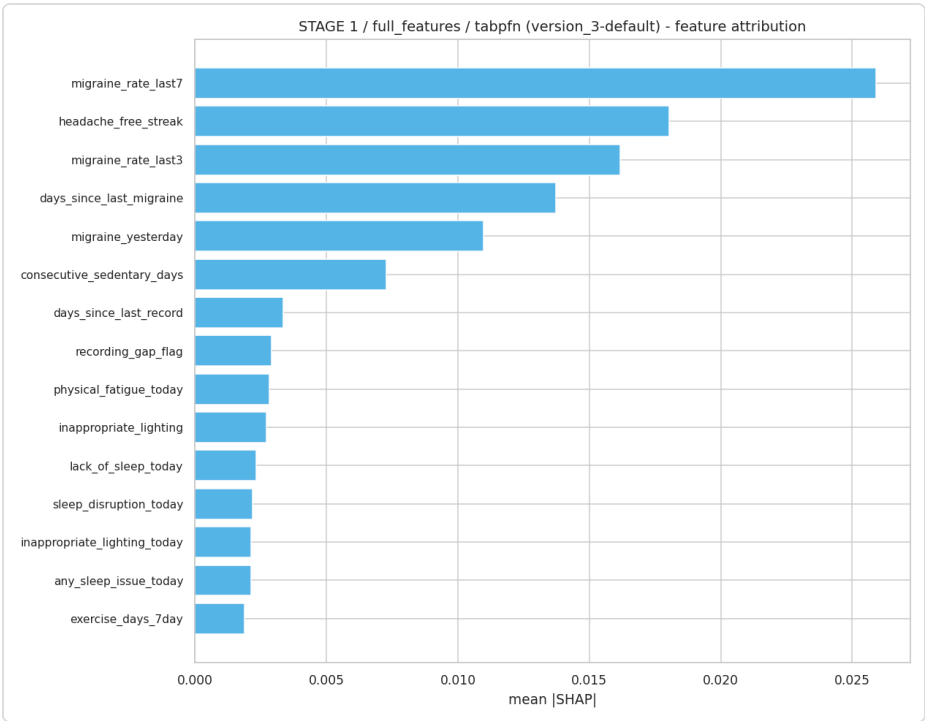
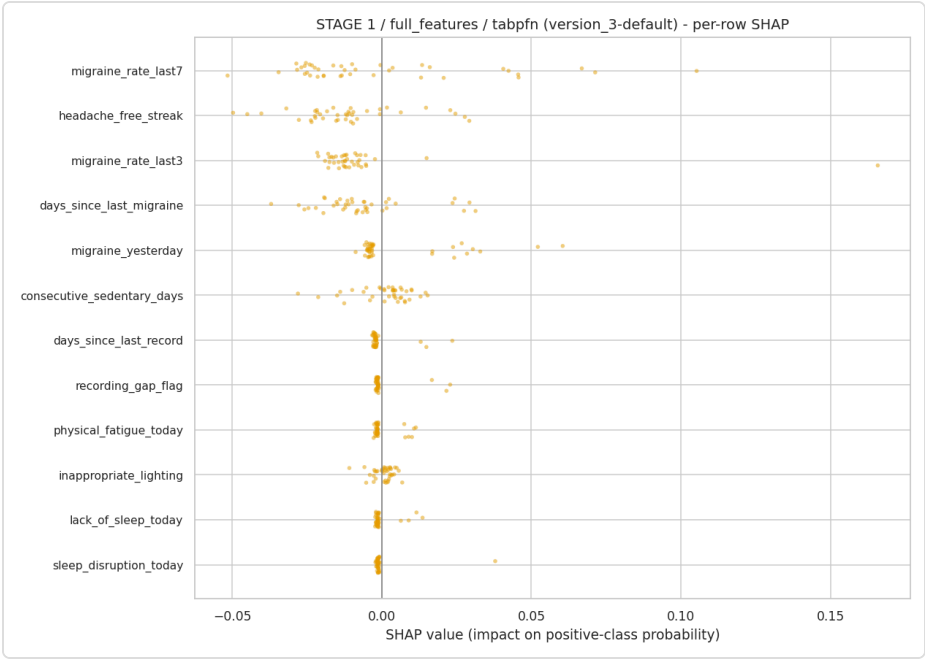
Spearman rank correlation over all 52 shared features: **+0.643** - moderate cross-architecture agreement on feature ordering.

rank	headline (tabpfn)	runner-up (xgboost)
1	migraine_rate_last7	migraine_rate_last7
2	headache_free_streak	headache_free_streak
3	migraine_rate_last3	consecutive_sedentary_days
4	days_since_last_migraine	exercise_days_7day
5	migraine_yesterday	migraine_rate_last3
6	consecutive_sedentary_days	days_since_last_migraine
7	days_since_last_record	dow
8	recording_gap_flag	days_since_last_record
9	physical_fatigue_today	consecutive_exercise_days
10	inappropriate_lighting	lack_of_sleep_today

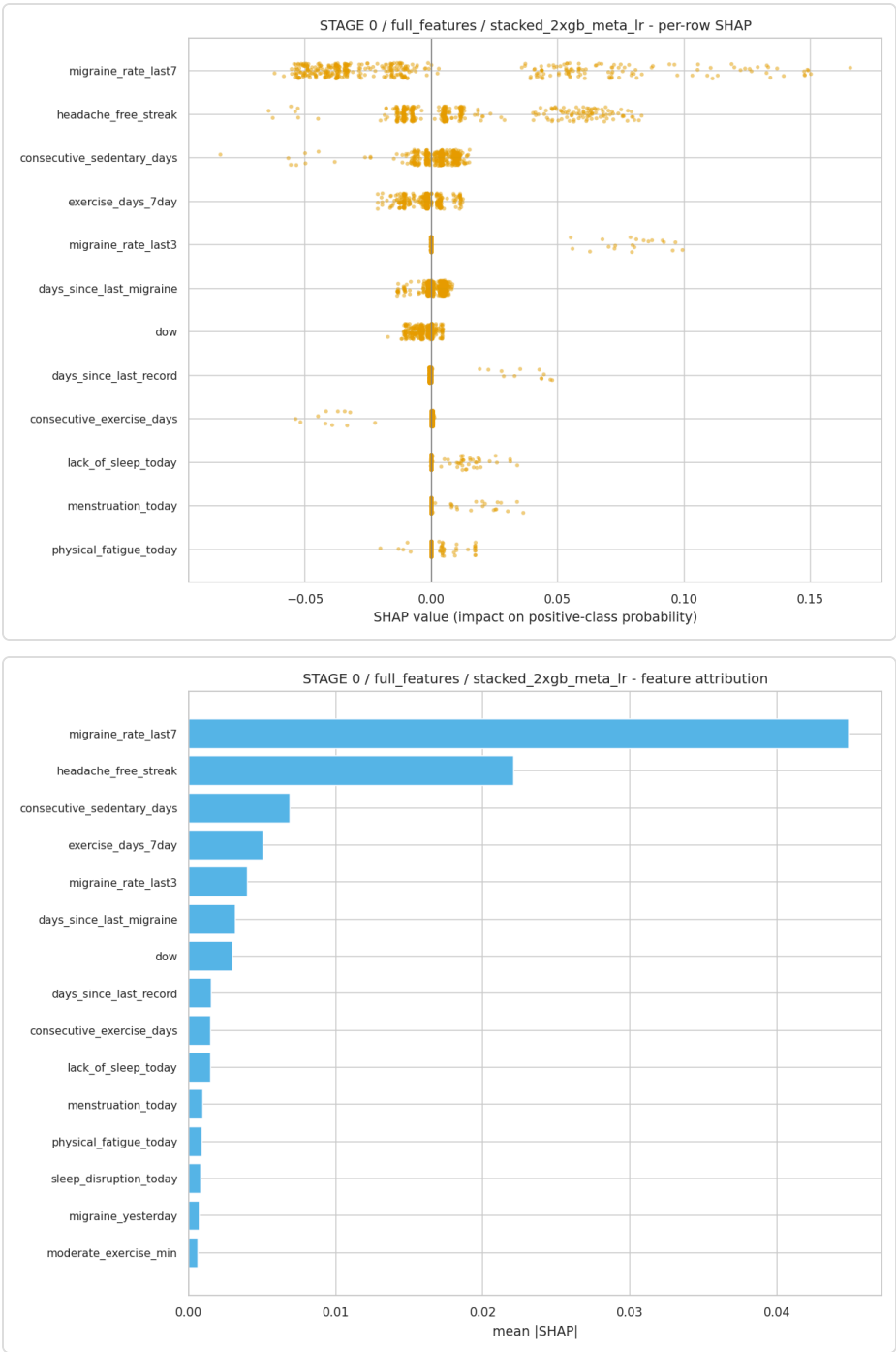
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (80_20/stratified)



runner-up: xgboost stacked_2xgb_meta_lr (70_15_15/stratified)



headache / no_rolling_features

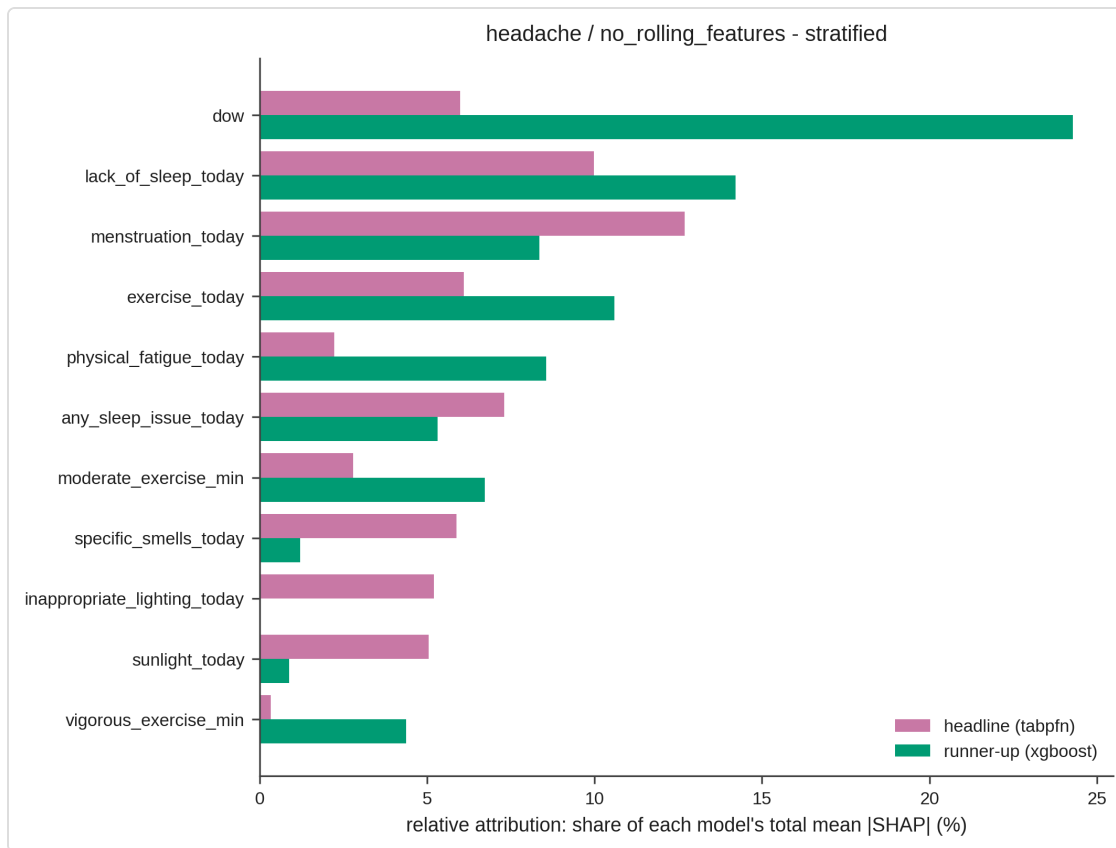
headline: **tabpfn/version_2-5-real** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.573, AUPRC 0.311, calib +0.81
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.577, AUPRC 0.296, calib +1.95

Spearman rank correlation over all 26 shared features: **+0.377** - weak cross-architecture agreement on feature ordering.

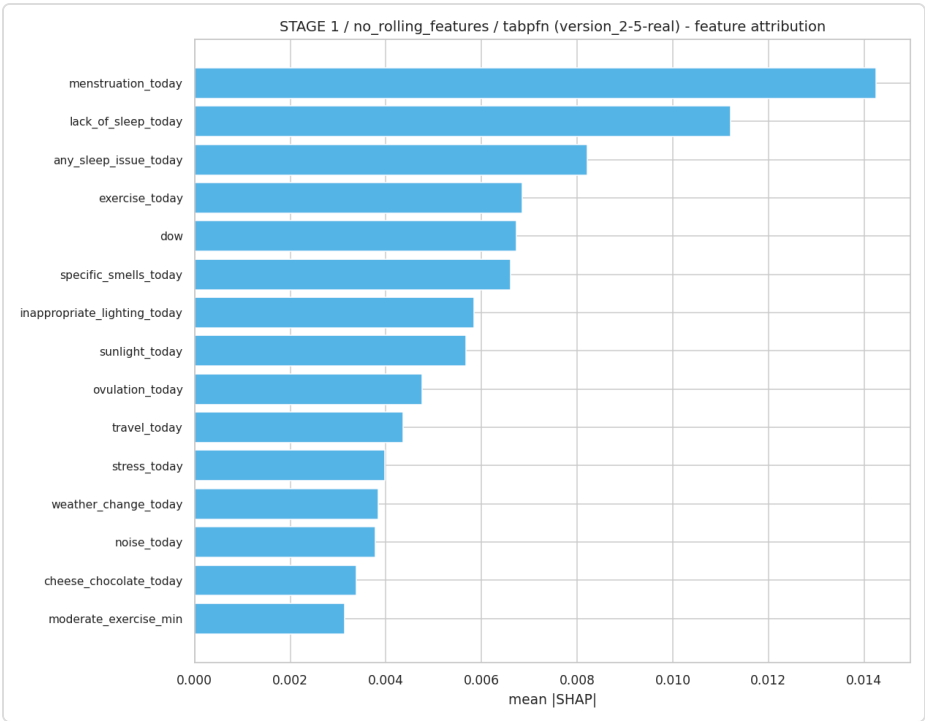
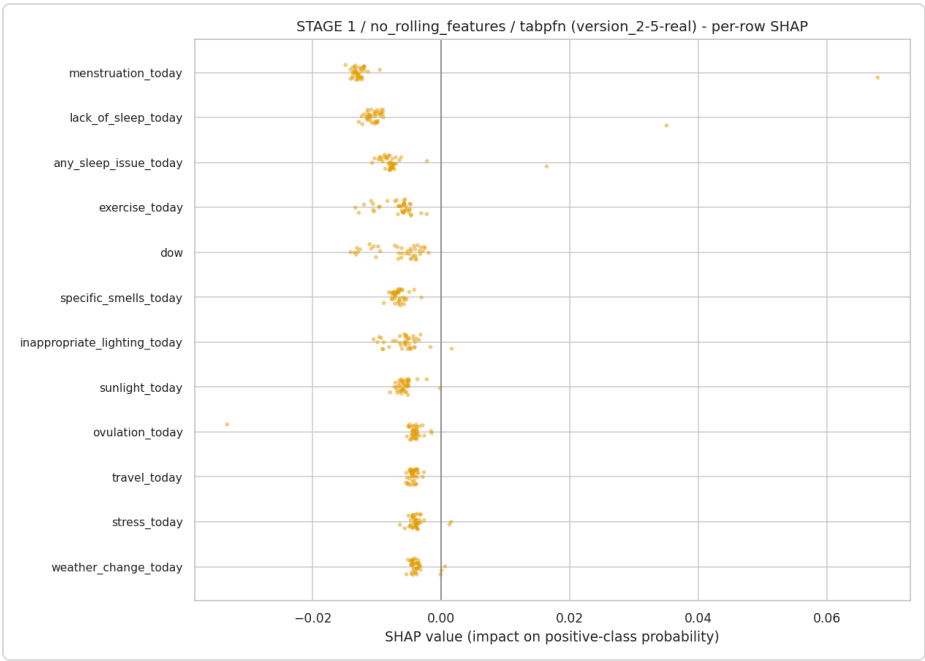
rank	headline (tabpfn)	runner-up (xgboost)
1	menstruation_today	dow

2	lack_of_sleep_today	lack_of_sleep_today
3	any_sleep_issue_today	exercise_today
4	exercise_today	physical_fatigue_today
5	dow	menstruation_today
6	specific_smells_today	moderate_exercise_min
7	inappropriate_lighting_today	any_sleep_issue_today
8	sunlight_today	vigorous_exercise_min
9	ovulation_today	ovulation_today
10	travel_today	stress_today

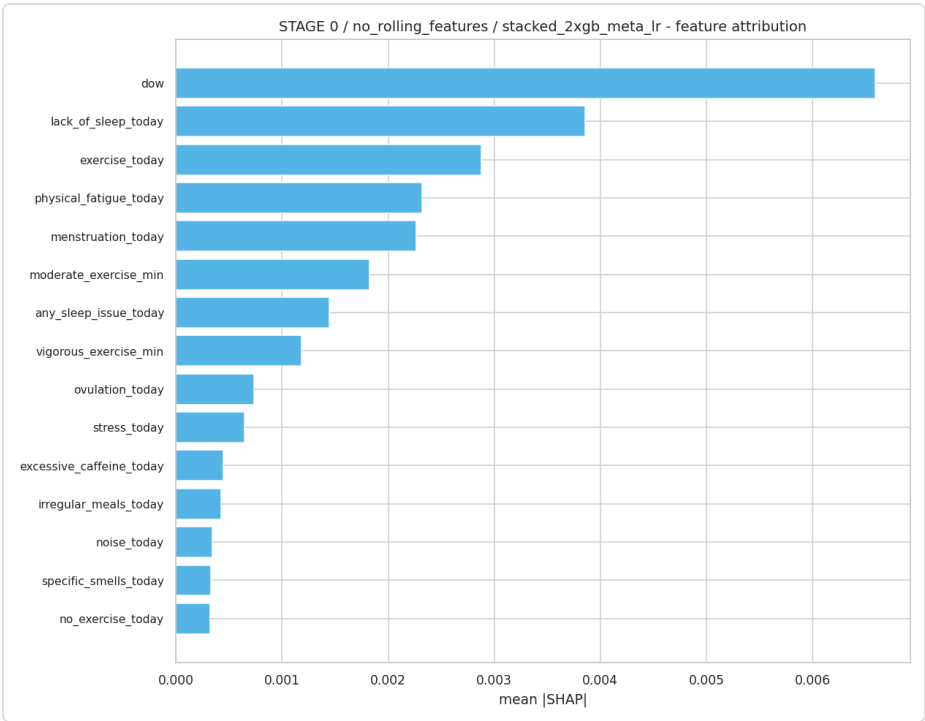
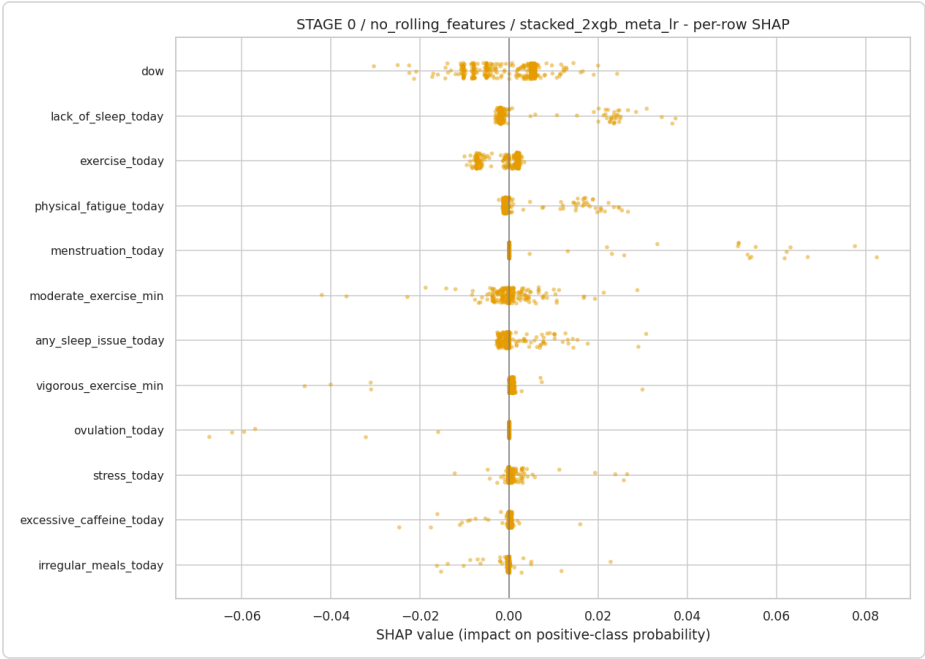
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_15_15/stratified)



runner-up: xgboost stacked_2xgb_meta_lr (70_15_15/stratified)



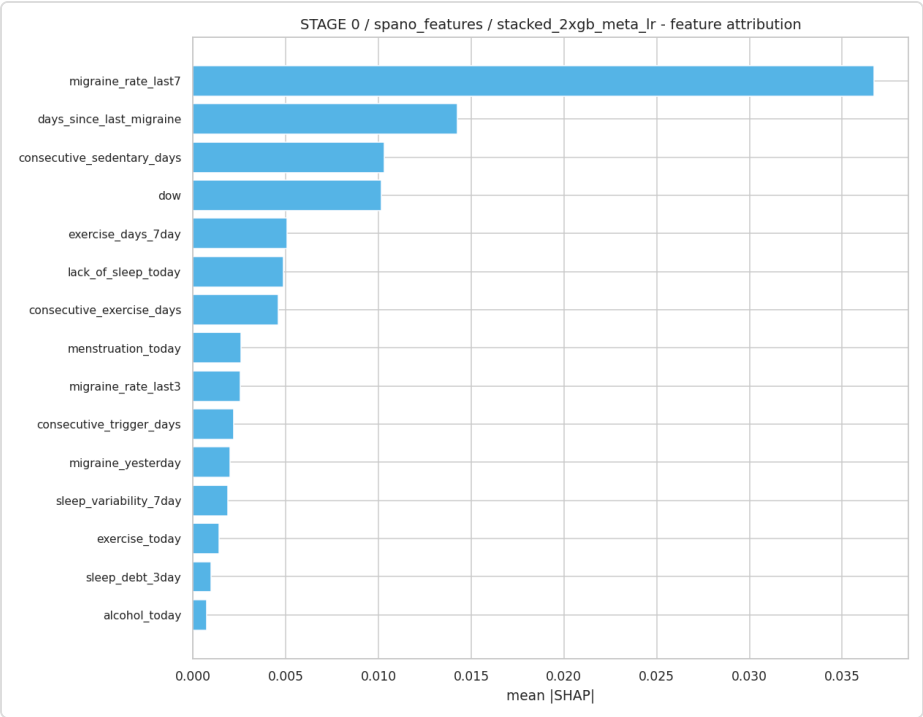
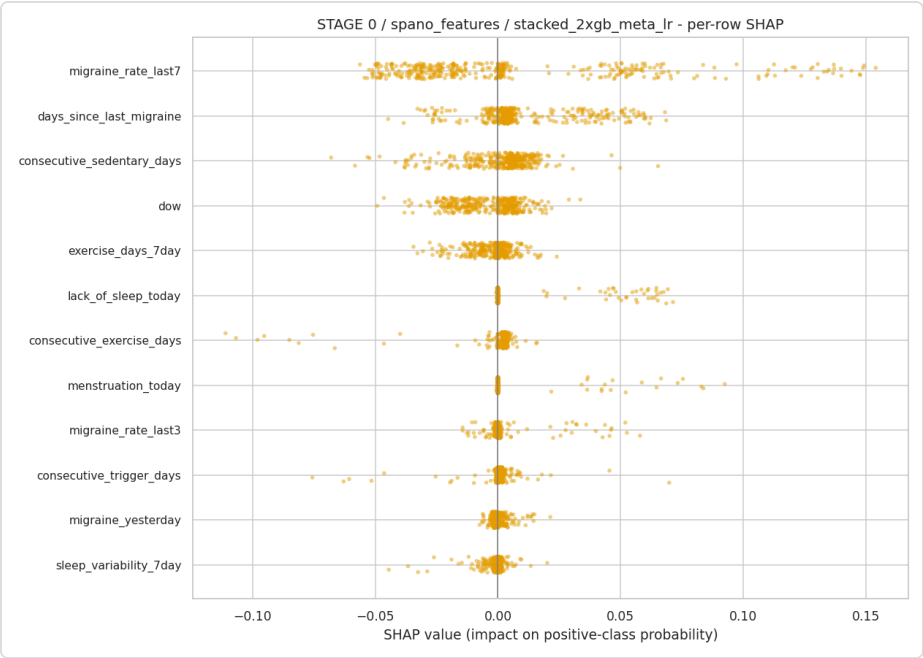
headache / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.664, AUPRC 0.413, calib +1.38. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	migraine_rate_last7	0.0367
2	days_since_last_migraine	0.0143
3	consecutive_sedentary_days	0.0103
4	dow	0.0102

5	exercise_days_7day	0.0051
6	lack_of_sleep_today	0.0049
7	consecutive_exercise_days	0.0046
8	menstruation_today	0.0026
9	migraine_rate_last3	0.0025
10	consecutive_trigger_days	0.0022

leaf: xgboost stacked_2xgb_meta_lr (70_15_15/stratified)



migraine / full_features

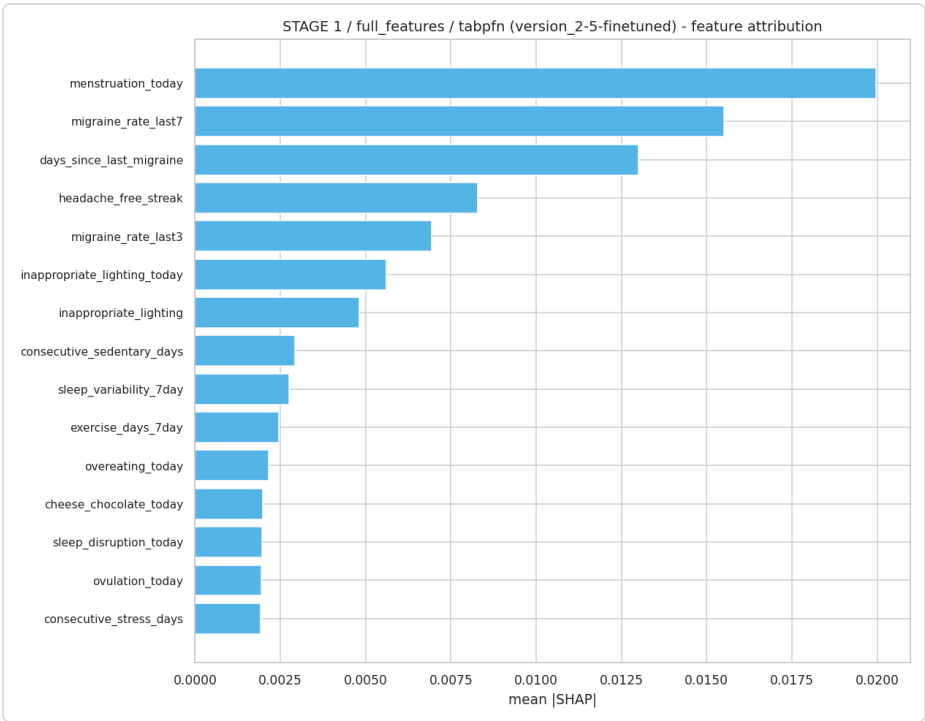
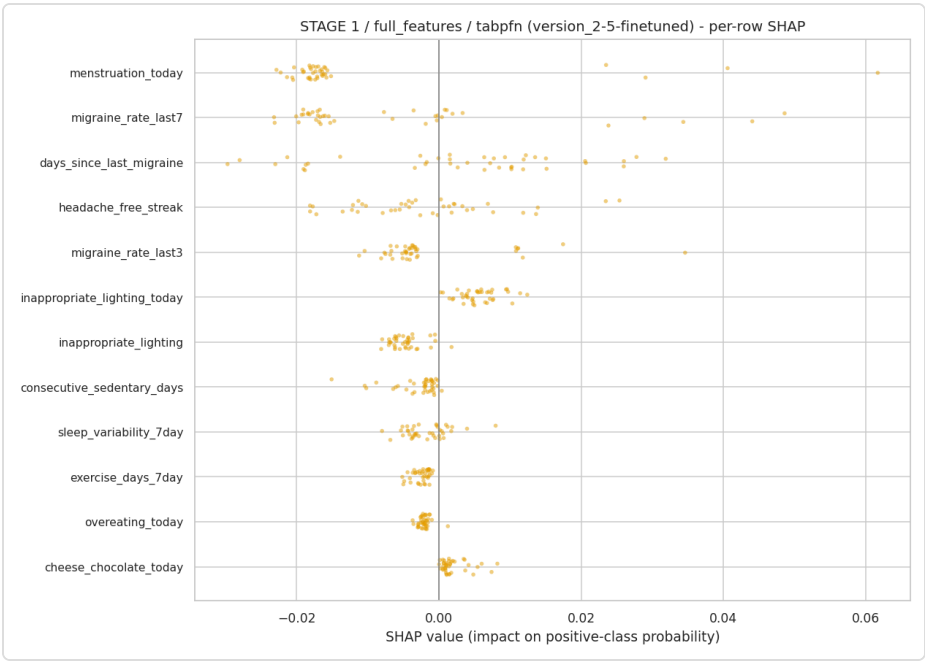
headline: **tabpfn/version_2-5-finetuned** [NonHP (library defaults)] | split 70_15_15/stratified |
hold-out AUROC 0.736, AUPRC 0.266, calib +1.01

runner-up: **tabpfn/version_2-5-auto** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.733, AUPRC 0.240, calib +0.77

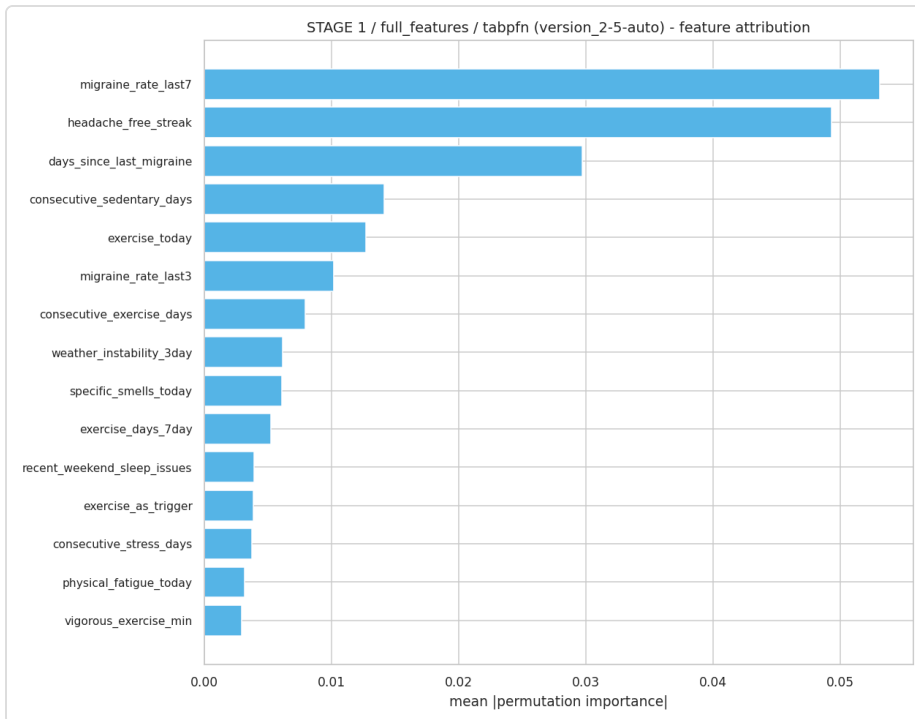
Spearman rank correlation over all 52 shared features: **+0.111** - weak cross-architecture agreement on feature ordering.

rank	headline (tabpfn)	runner-up (autotabpfn)
1	menstruation_today	migraine_rate_last7
2	migraine_rate_last7	headache_free_streak
3	days_since_last_migraine	days_since_last_migraine
4	headache_free_streak	consecutive_sedentary_days
5	migraine_rate_last3	exercise_today
6	inappropriate_lighting_today	migraine_rate_last3
7	inappropriate_lighting	consecutive_exercise_days
8	consecutive_sedentary_days	weather_instability_3day
9	sleep_variability_7day	specific_smells_today
10	exercise_days_7day	exercise_days_7day

headline: tabpfn tabpfn (70_15_15/stratified)



runner-up: autotabpfn tabpfn (70_15_15/stratified)



migraine / no_rolling_features

headline: **tabpfn/version_2-5-finetuned** [NonHP (library defaults)] | split 70_30/stratified | hold-out AUROC 0.614, AUPRC 0.128, calib +0.57

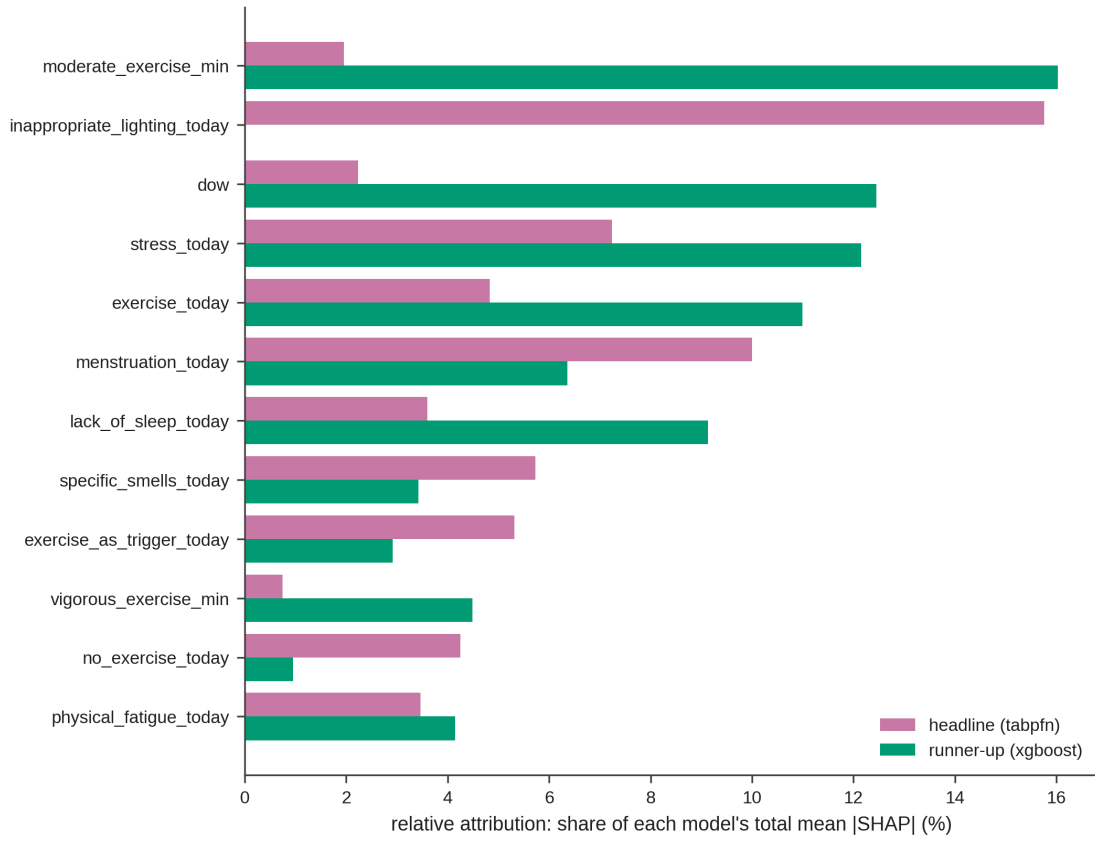
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.549, AUPRC 0.109, calib +0.57

Spearman rank correlation over all 26 shared features: **+0.125** - weak cross-architecture agreement on feature ordering.

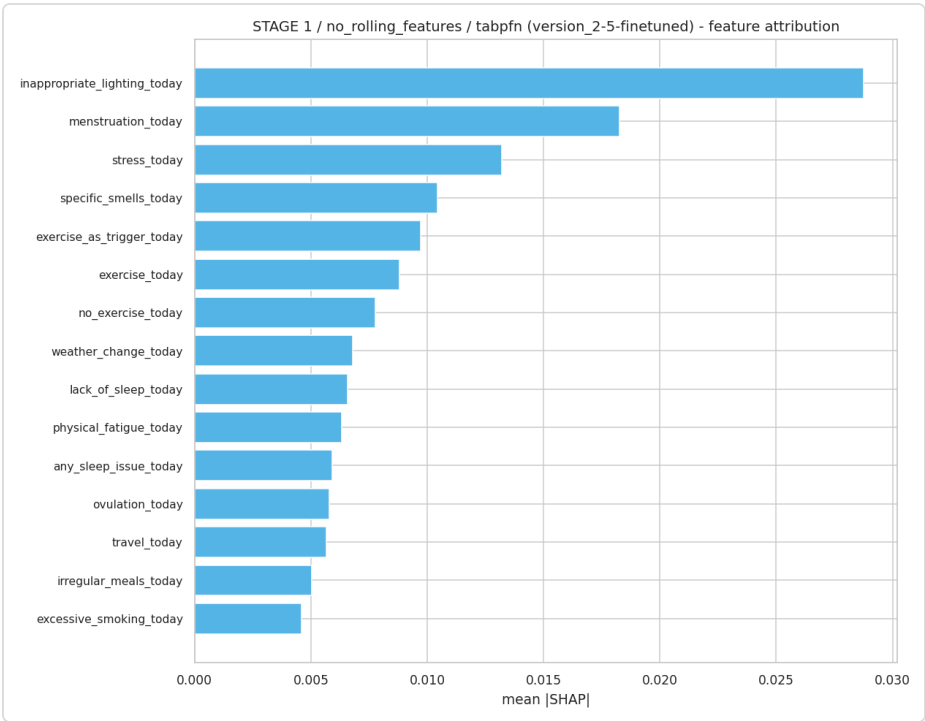
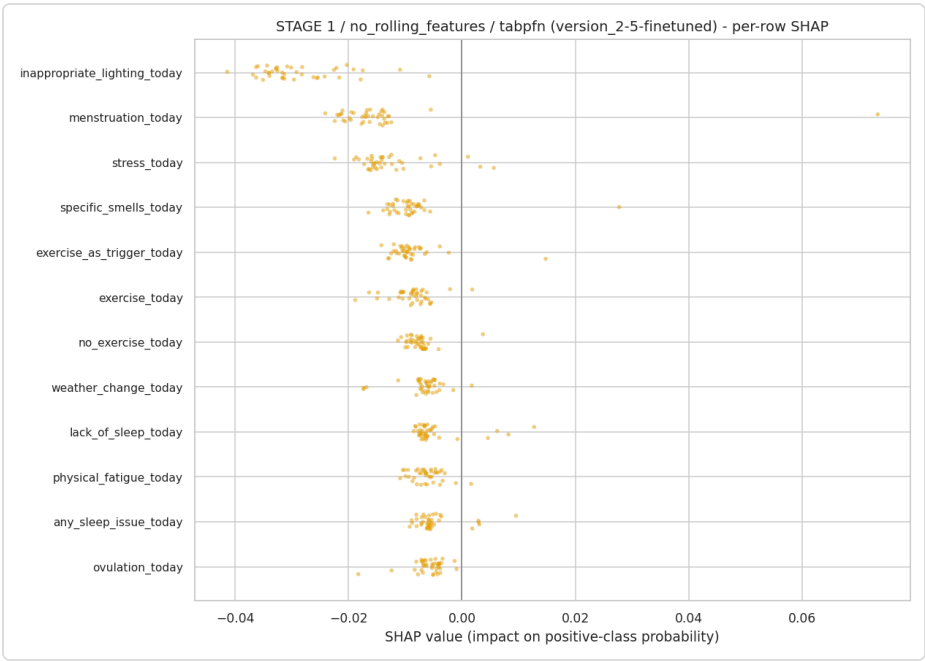
rank	headline (tabpfn)	runner-up (xgboost)
1	inappropriate_lighting_today	moderate_exercise_min
2	menstruation_today	dow
3	stress_today	stress_today
4	specific_smells_today	exercise_today
5	exercise_as_trigger_today	lack_of_sleep_today
6	exercise_today	menstruation_today
7	no_exercise_today	vigorous_exercise_min
8	weather_change_today	physical_fatigue_today
9	lack_of_sleep_today	emotional_changes_today
10	physical_fatigue_today	specific_smells_today

Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.

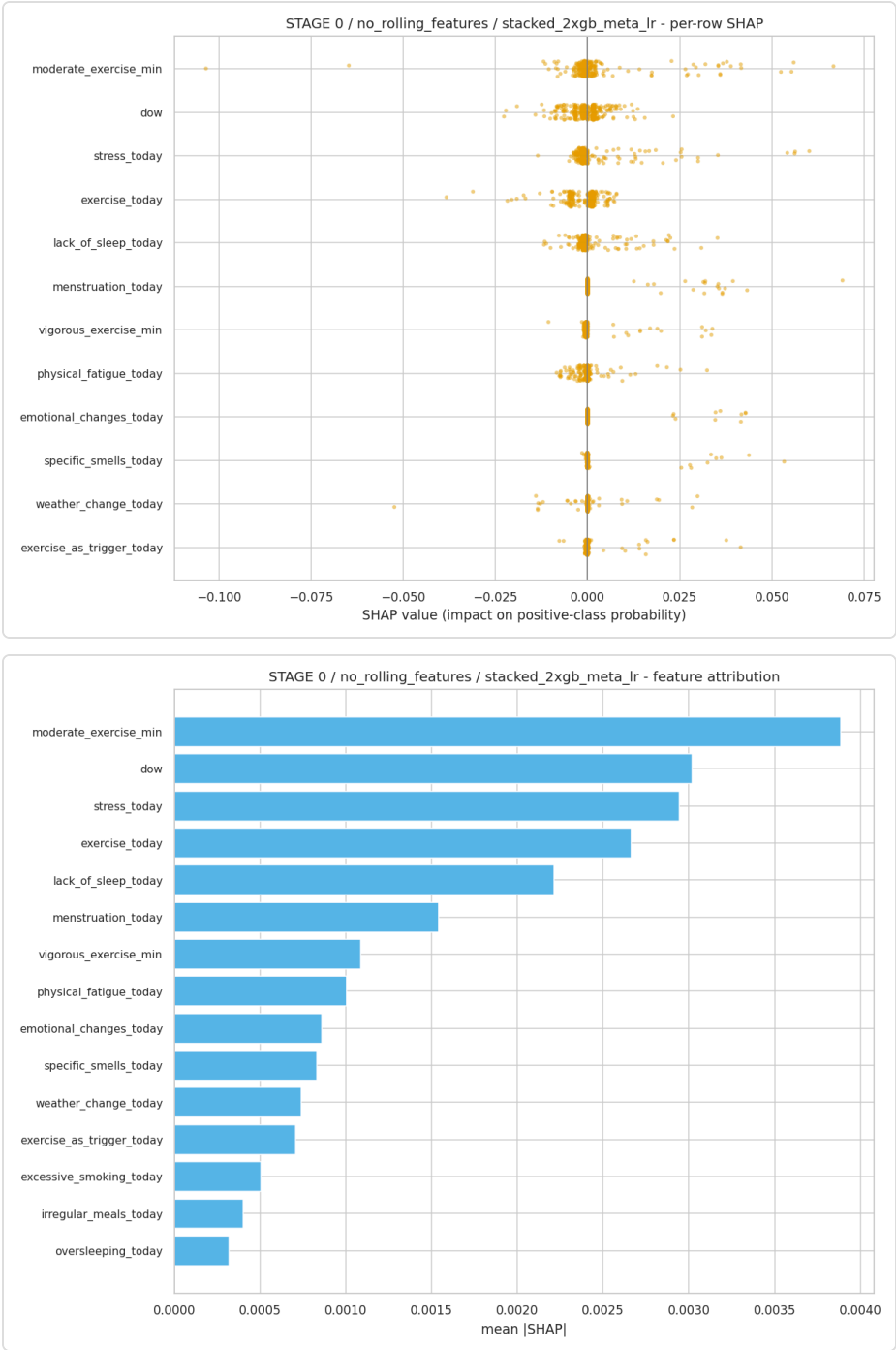
migraine / no_rolling_features - stratified



headline: tabpfn tabpfn (70_30/stratified)



runner-up: xgboost stacked_2xgb_meta_lr (70_15_15/stratified)



migraine / park_features

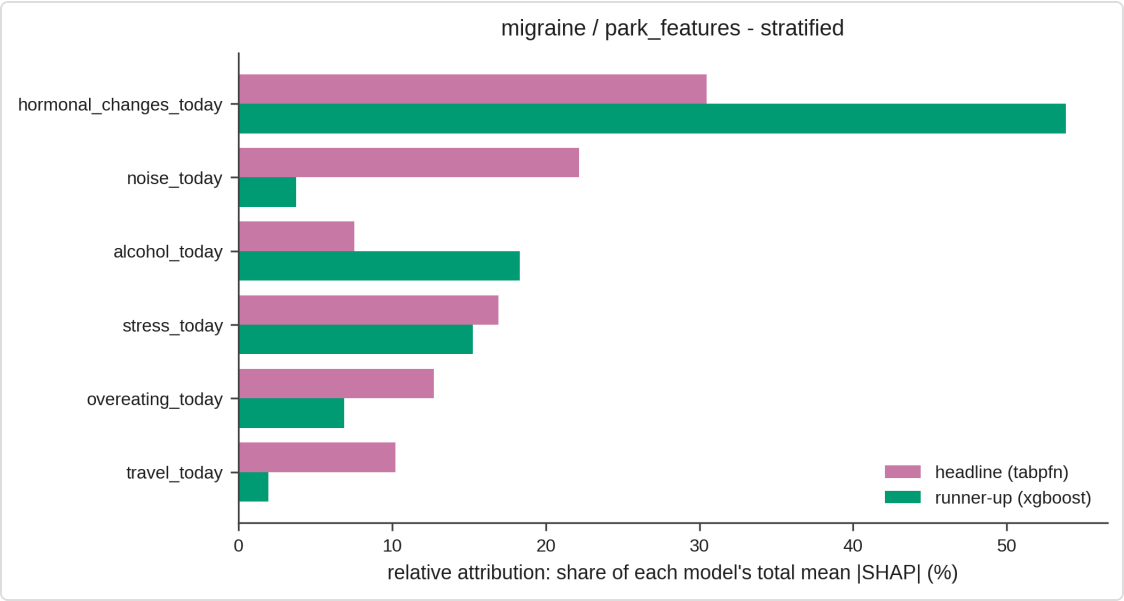
headline: **tabpfn/version_3-default** [NonHP (library defaults)] | split 80_20/stratified | hold-out AUROC 0.618, AUPRC 0.119, calib +0.59
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/stratified | hold-out AUROC 0.576, AUPRC 0.117, calib +0.56

Spearman rank correlation over all 6 shared features: **+0.257** - weak cross-architecture agreement on feature ordering.

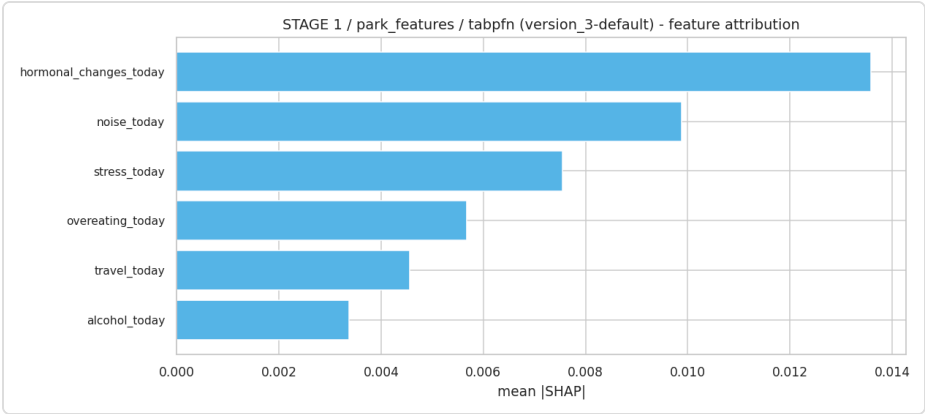
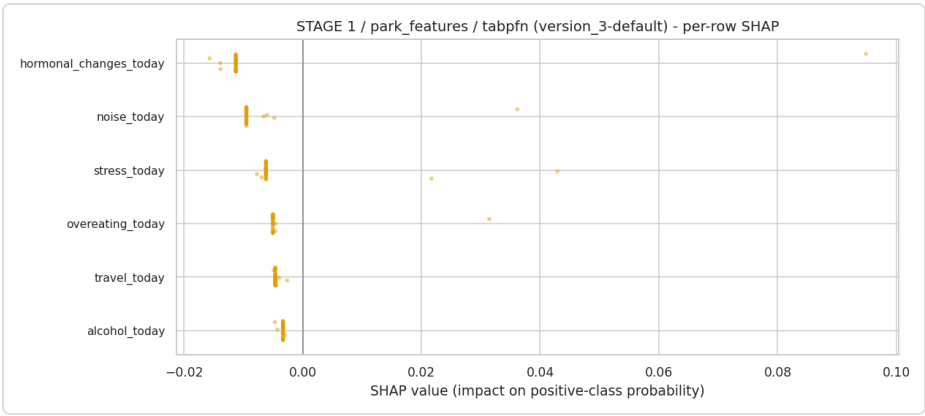
rank	headline (tabpfn)	runner-up (xgboost)
1	hormonal_changes_today	hormonal_changes_today

2	noise_today	alcohol_today
3	stress_today	stress_today
4	overeating_today	overeating_today
5	travel_today	noise_today
6	alcohol_today	travel_today
7		
8		
9		
10		

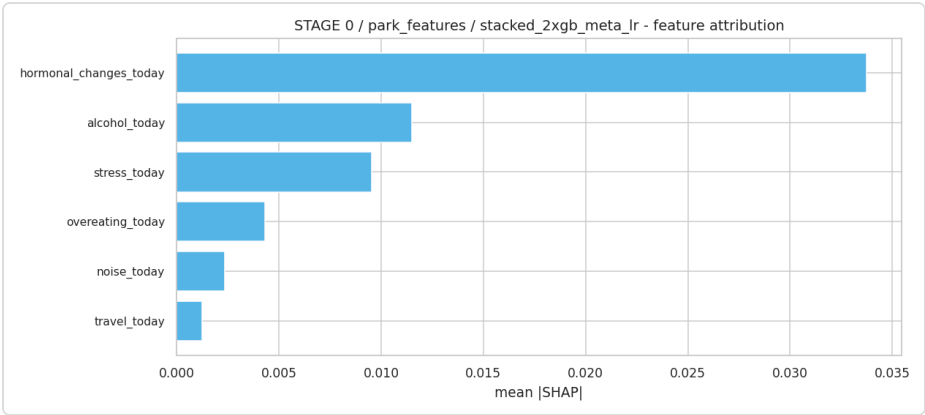
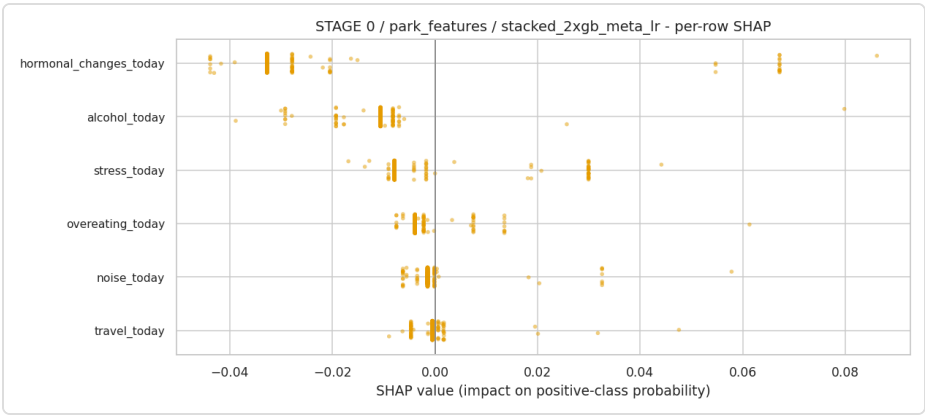
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (80_20/stratified)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/stratified)

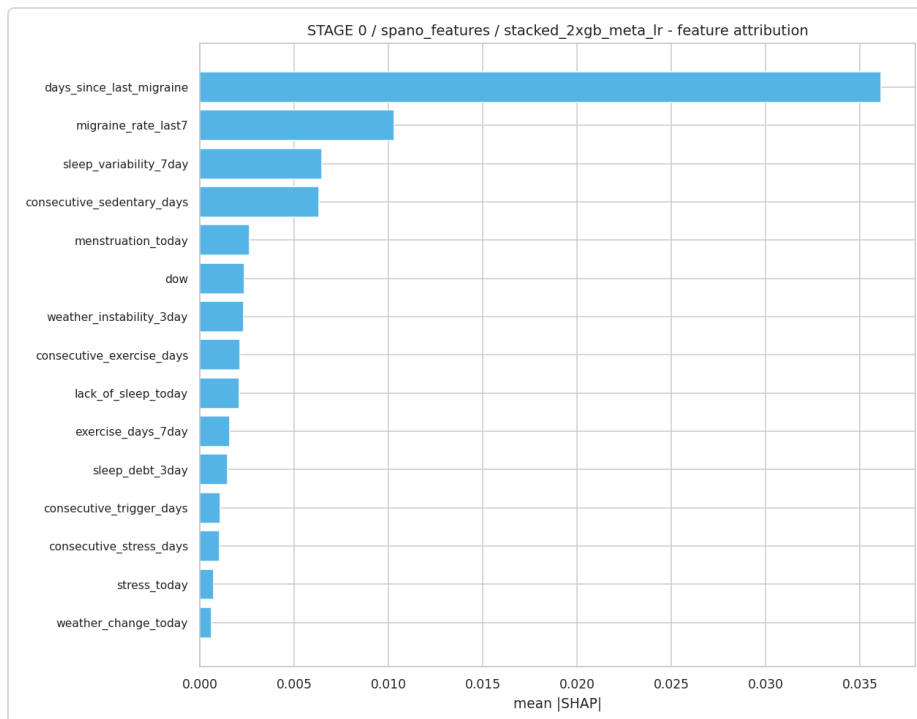
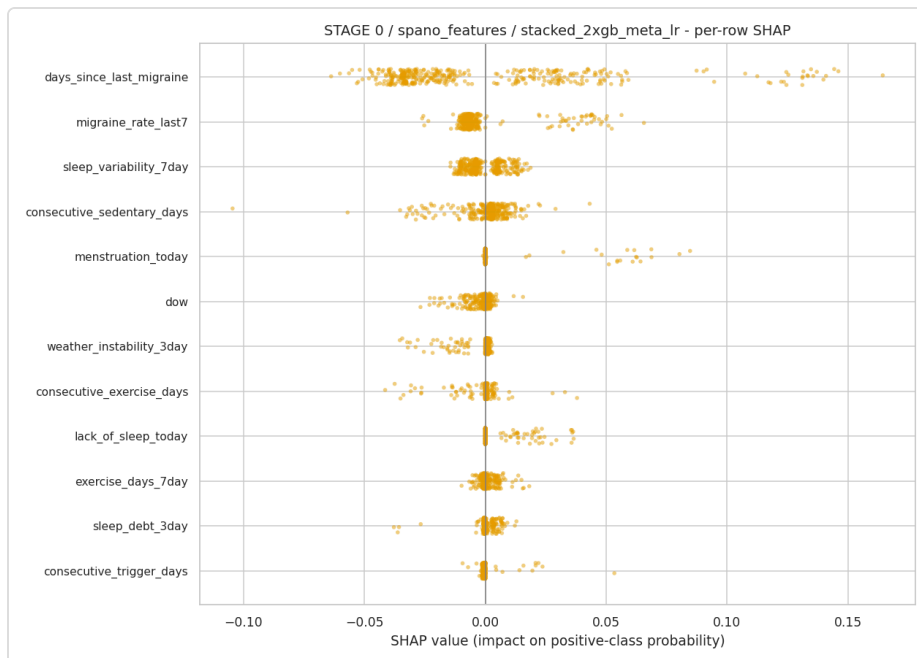


migraine / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_15_15/stratified | hold-out AUROC 0.716, AUPRC 0.230, calib +0.94. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	days_since_last_migraine	0.0361
2	migraine_rate_last7	0.0103
3	sleep_variability_7day	0.0065
4	consecutive_sedentary_days	0.0063
5	menstruation_today	0.0026
6	dow	0.0024
7	weather_instability_3day	0.0023
8	consecutive_exercise_days	0.0021
9	lack_of_sleep_today	0.0021
10	exercise_days_7day	0.0016

leaf: xgboost stacked_2xgb_meta_lr (70_15_15/stratified)



Patient hold-out (generalisation to unseen patients)

headache / full_features

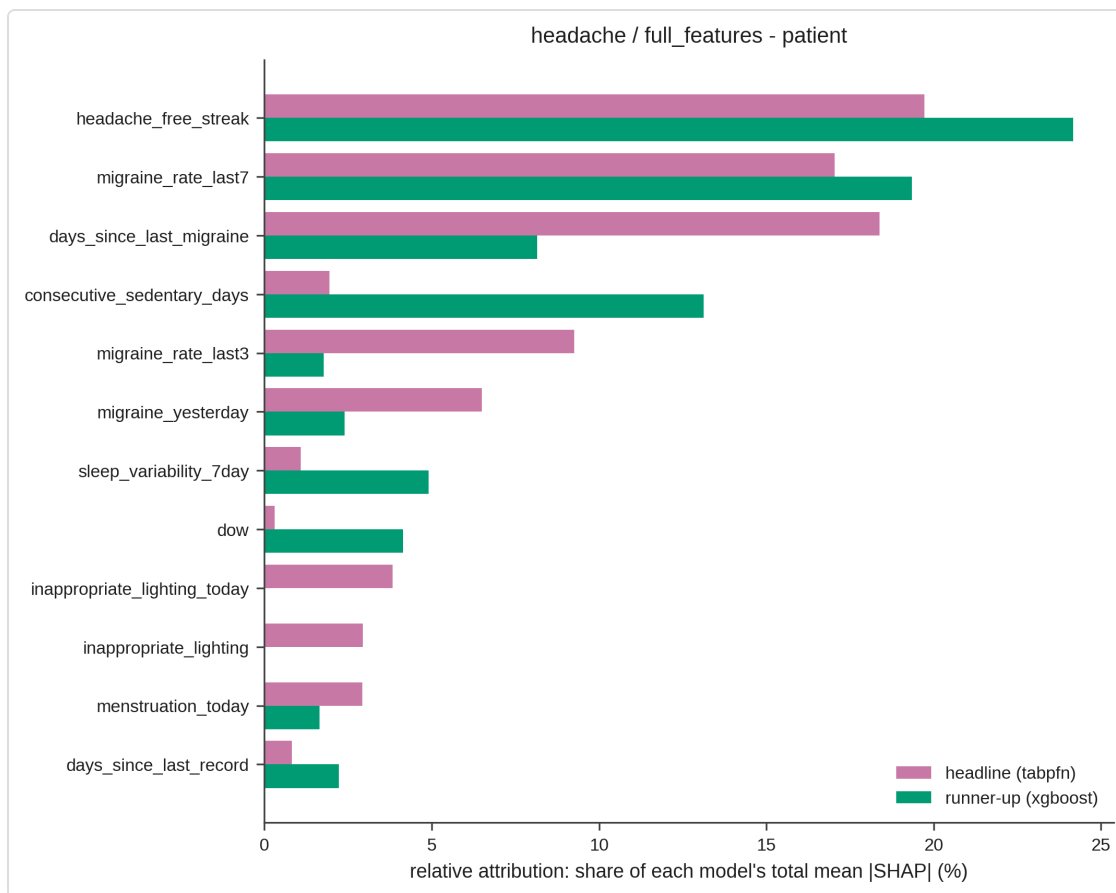
headline: **tabpfn/version_3-default** [NonHP (library defaults)] | split 70_15_15/patient | hold-out AUROC 0.646, AUPRC 0.391, calib +0.90

runner-up: **stacked_2xgb_meta_lr** [single_AUROC/HP020] | split 80_20/patient | hold-out AUROC 0.638, AUPRC 0.381, calib +1.17

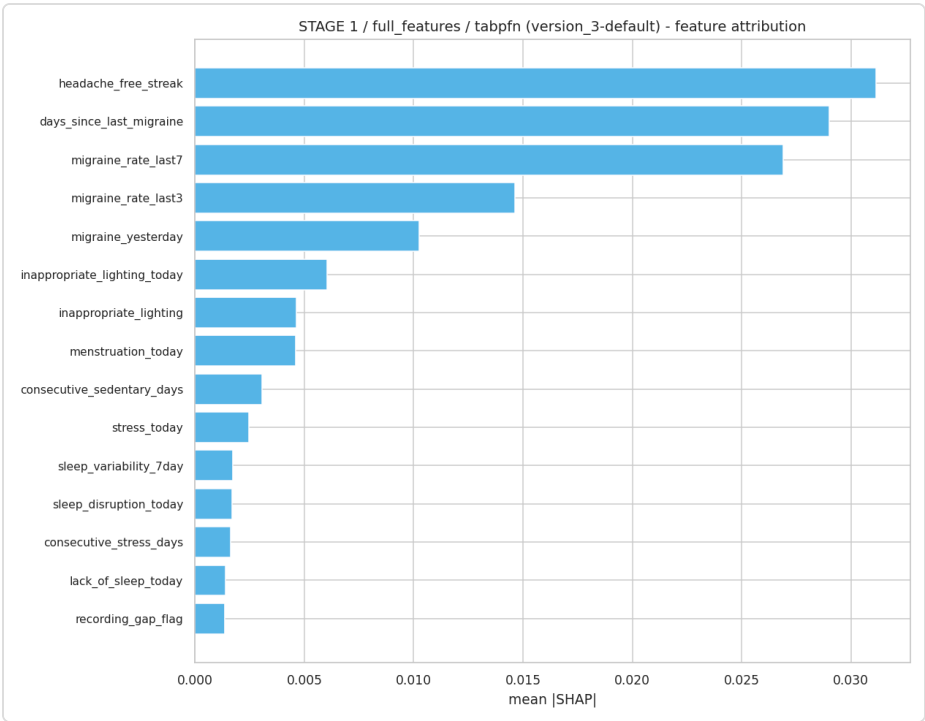
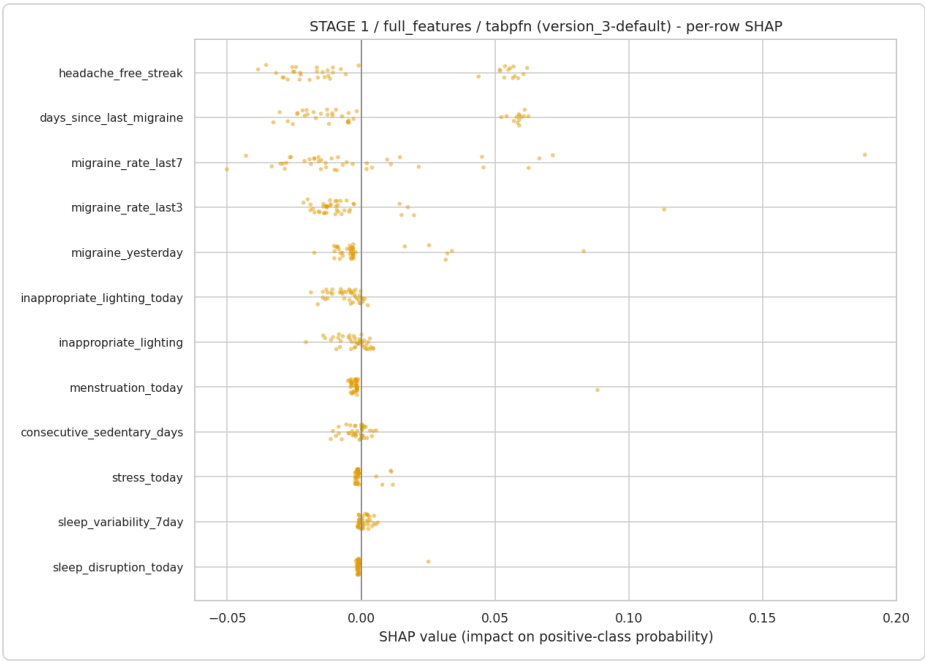
Spearman rank correlation over all 52 shared features: **+0.592** - moderate cross-architecture agreement on feature ordering.

rank	headline (tabpfn)	runner-up (xgboost)
1	headache_free_streak	headache_free_streak
2	days_since_last_migraine	migraine_rate_last7
3	migraine_rate_last7	consecutive_sedentary_days
4	migraine_rate_last3	days_since_last_migraine
5	migraine_yesterday	sleep_variability_7day
6	inappropriate_lighting_today	dow
7	inappropriate_lighting	migraine_yesterday
8	menstruation_today	days_since_last_record
9	consecutive_sedentary_days	migraine_rate_last3
10	stress_today	exercise_today

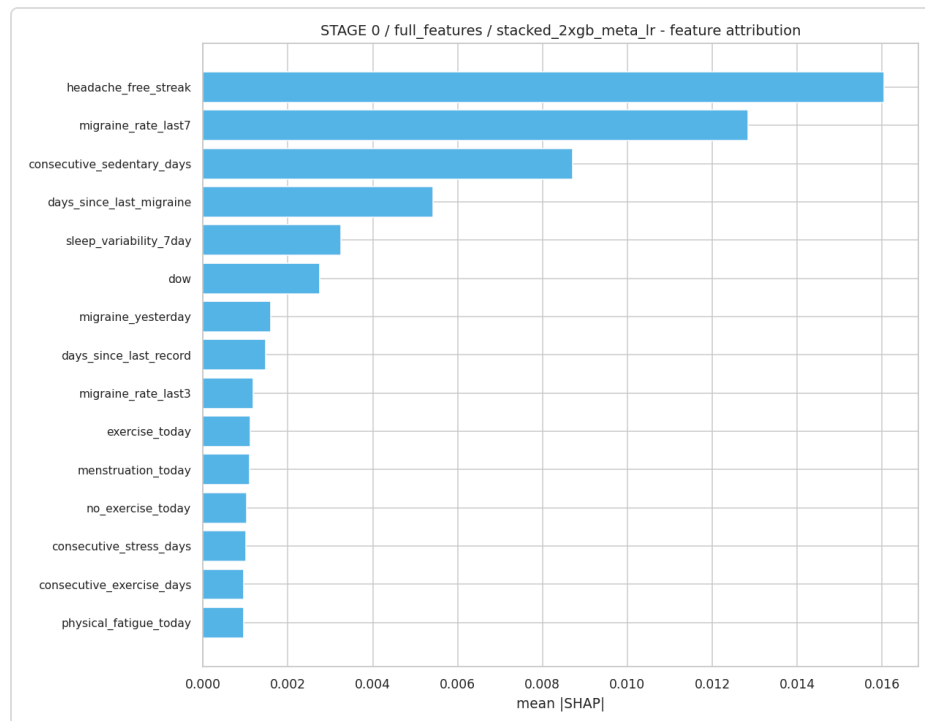
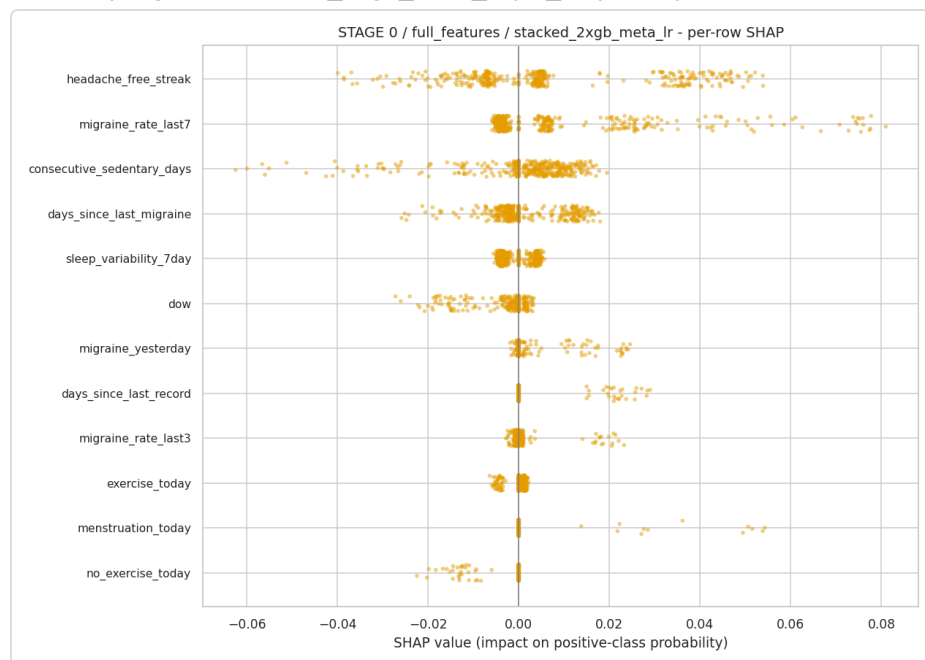
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_15_15/patient)



runner-up: xgboost stacked_2xgb_meta_lr (80_20/patient)



headache / no_rolling_features

This cell's headline is not significantly above chance (AUROC 95% CI [0.479-0.567] includes 0.5); its attributions describe a near-chance model and are not evidence of a real effect.

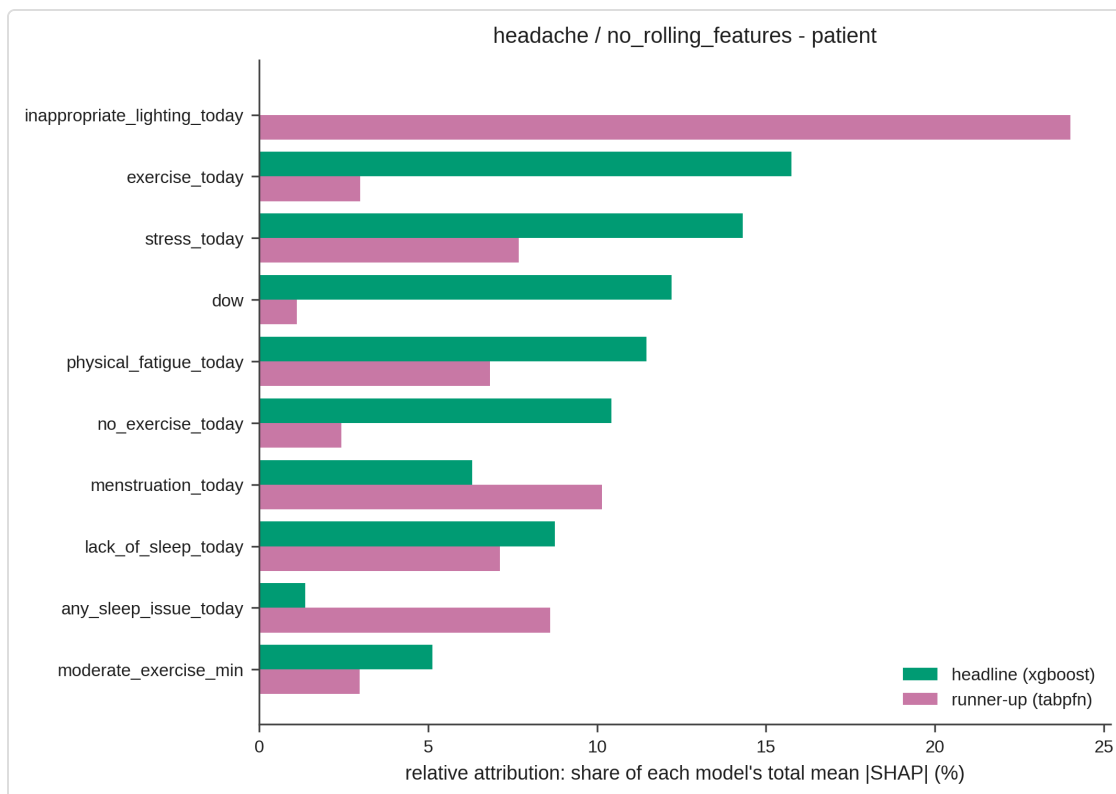
headline: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 80_20/patient | hold-out AUROC 0.522, AUPRC 0.264, calib +1.00

runner-up: **tabpfn/version_3-binary** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.486, AUPRC 0.237, calib -0.02

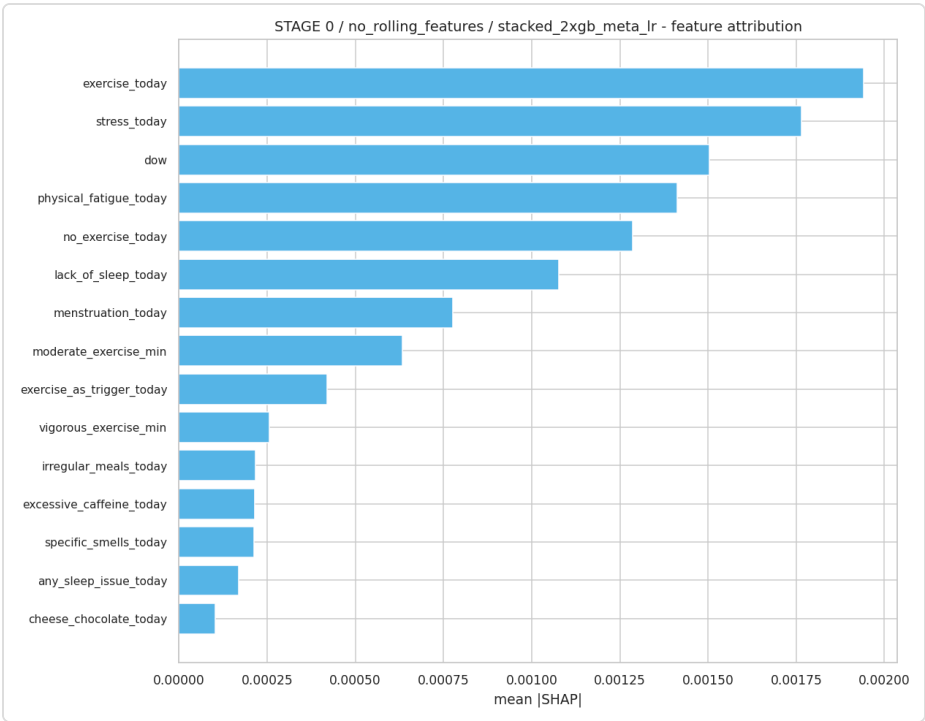
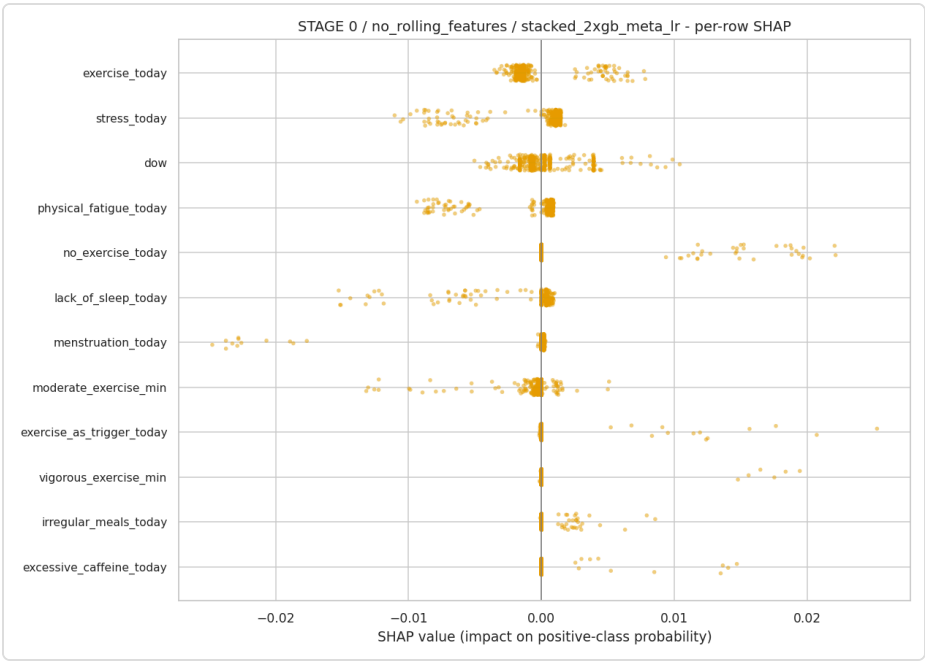
Spearman rank correlation over all 26 shared features: **+0.251** - weak cross-architecture agreement on feature ordering.

rank	headline (xgboost)	runner-up (tabpfn)
1	exercise_today	inappropriate_lighting_today
2	stress_today	menstruation_today
3	dow	any_sleep_issue_today
4	physical_fatigue_today	stress_today
5	no_exercise_today	lack_of_sleep_today
6	lack_of_sleep_today	physical_fatigue_today
7	menstruation_today	exercise_today
8	moderate_exercise_min	moderate_exercise_min
9	exercise_as_trigger_today	emotional_changes_today
10	vigorous_exercise_min	ovulation_today

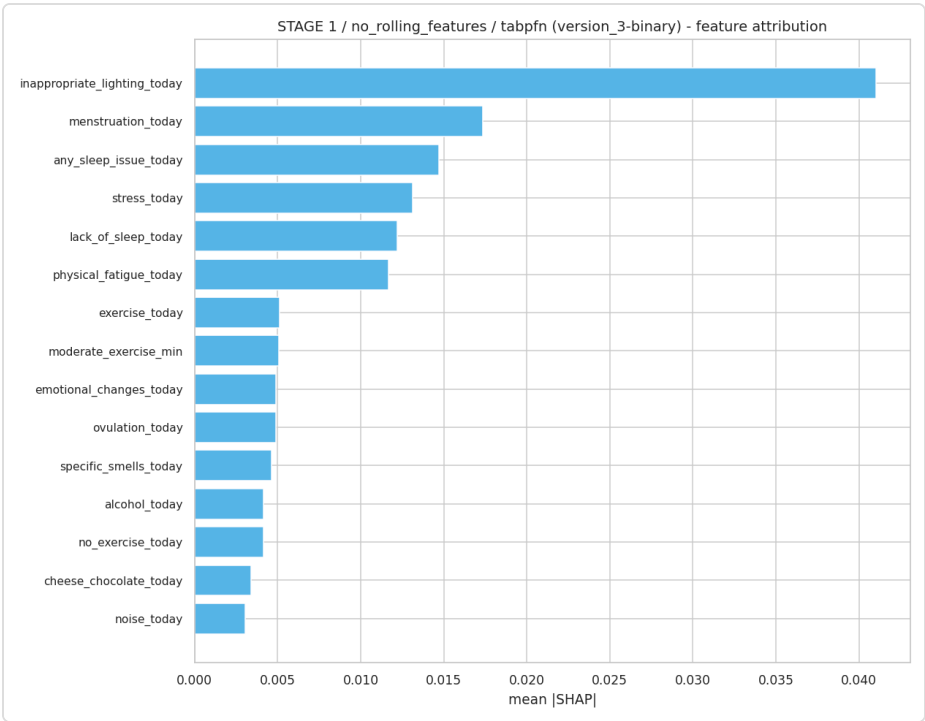
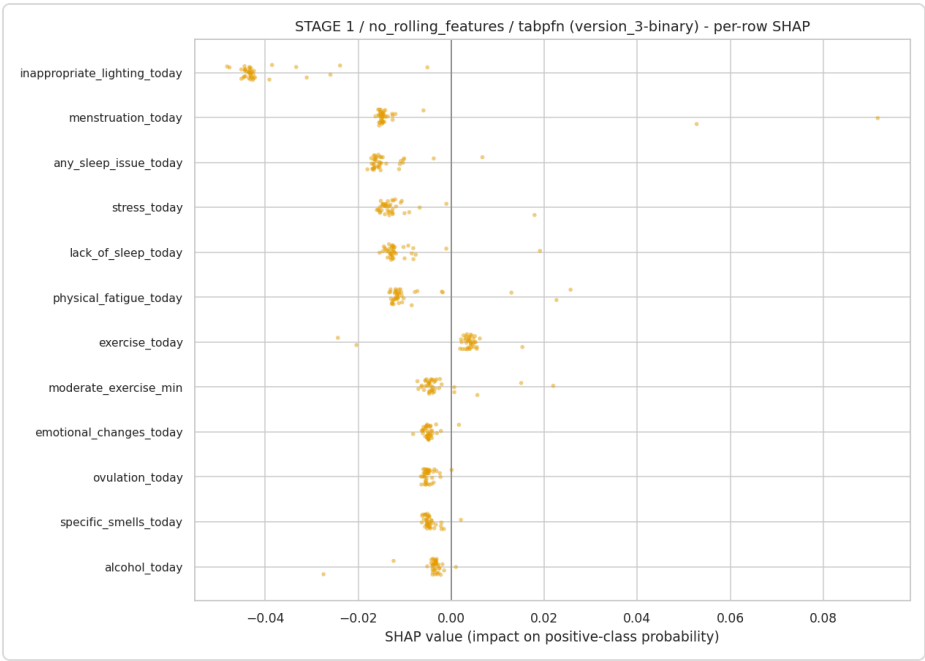
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: xgboost stacked_2xgb_meta_lr (80_20/patient)



runner-up: tabpfn tabpfn (70_30/patient)



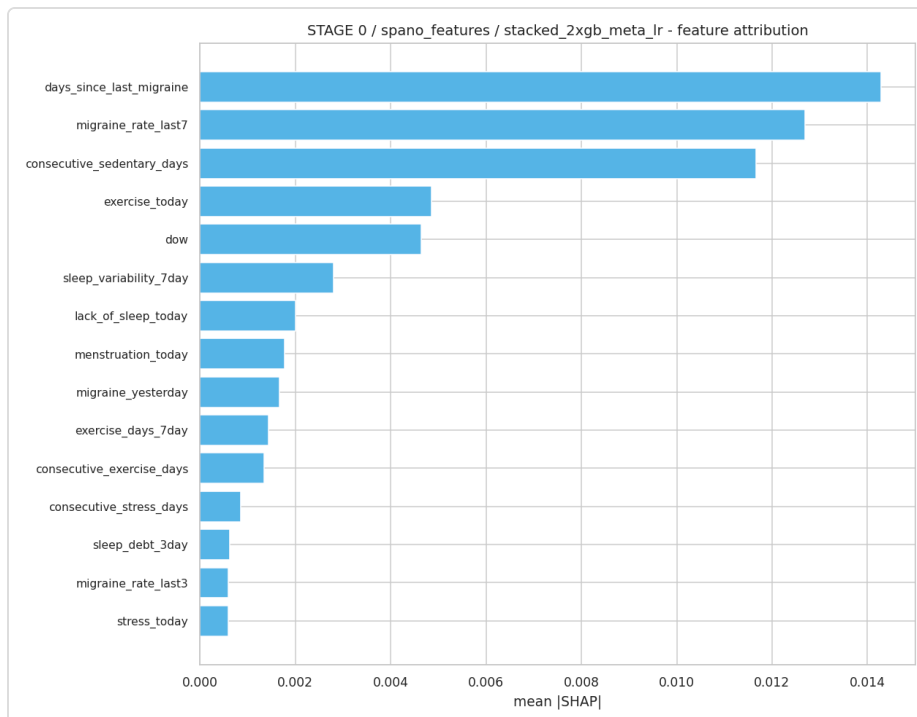
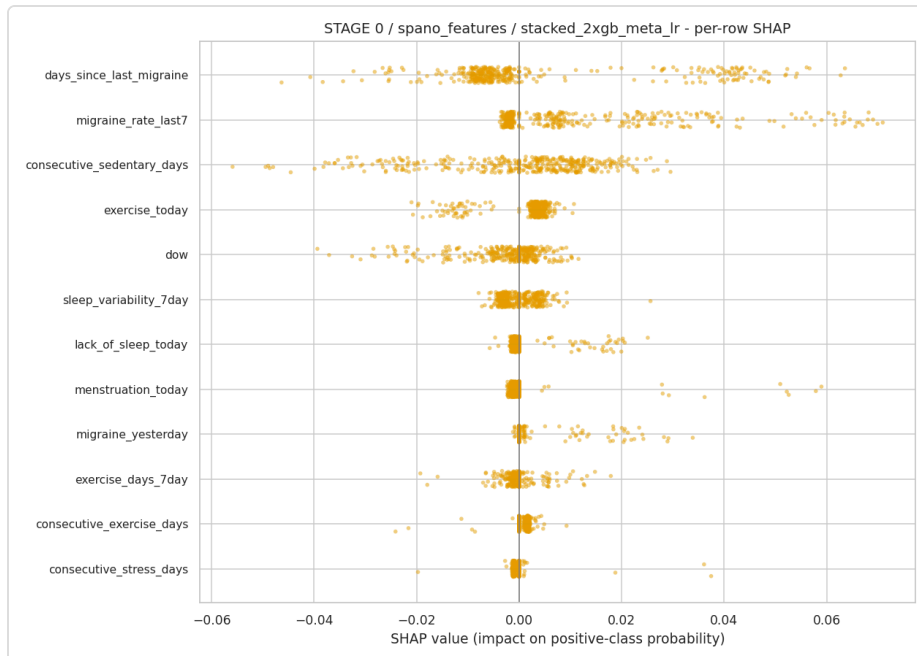
headache / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.631, AUPRC 0.354, calib +1.29. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	days_since_last_migraine	0.0143
2	migraine_rate_last7	0.0127
3	consecutive_sedentary_days	0.0117
4	exercise_today	0.0049

5	dow	0.0046
6	sleep_variability_7day	0.0028
7	lack_of_sleep_today	0.0020
8	menstruation_today	0.0018
9	migraine_yesterday	0.0017
10	exercise_days_7day	0.0014

leaf: xgboost stacked_2xgb_meta_lr (70_30/patient)



migraine / full_features

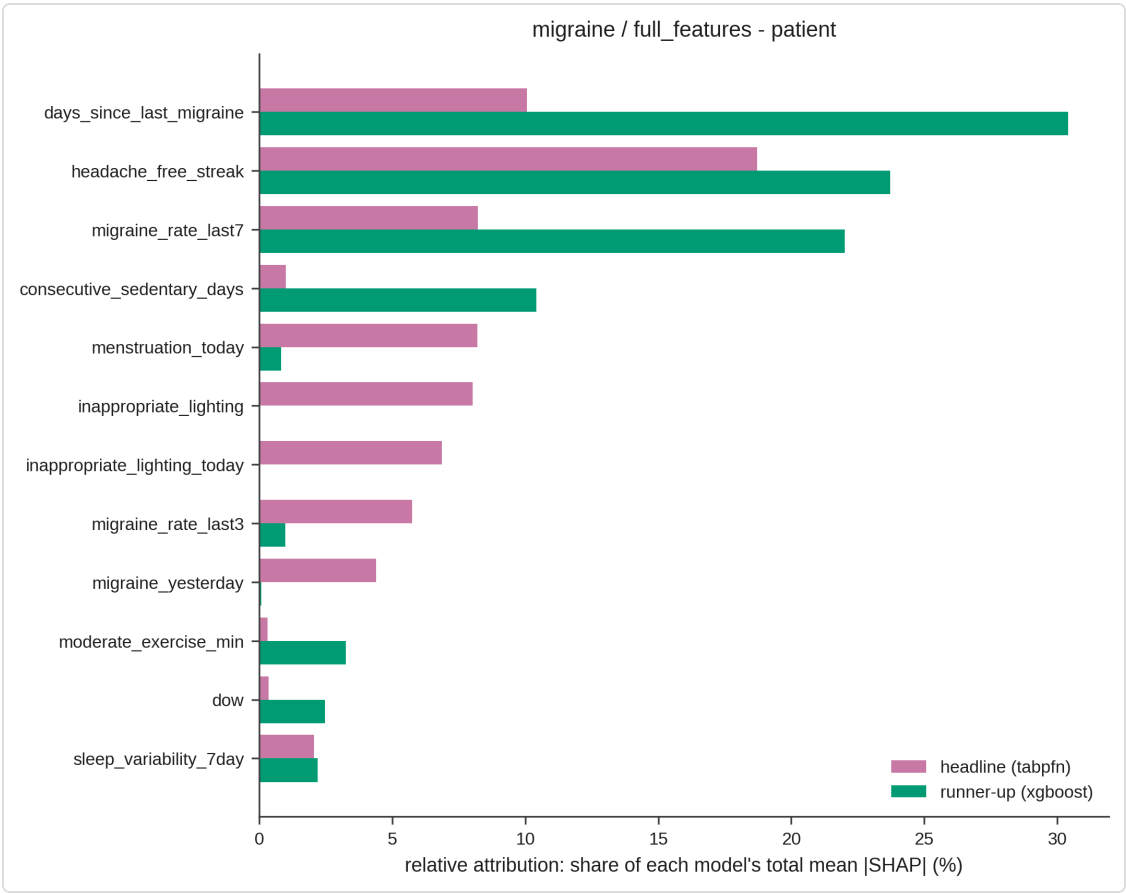
headline: **tabpfn/version_3-binary** [NonHP (library defaults)] | split 70_15_15/patient | hold-out AUROC 0.716, AUPRC 0.174, calib +0.71

runner-up: **stacked_2xgb_meta_lr** [pareto_AUROC_slope/slope_closest] | split 70_15_15/patient | hold-out AUROC 0.691, AUPRC 0.186, calib +1.18

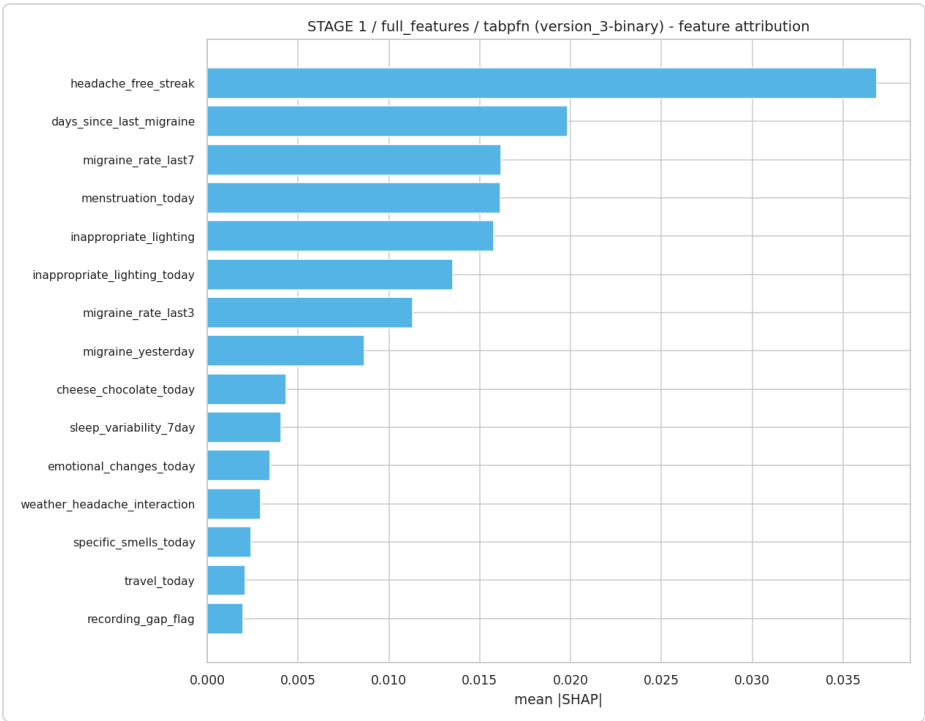
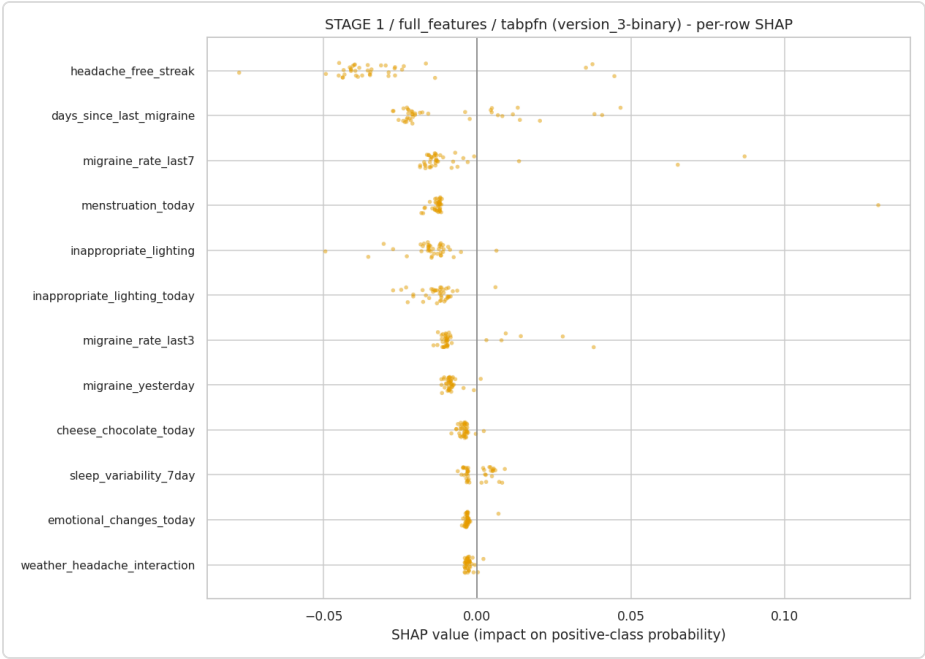
Spearman rank correlation over all 52 shared features: **+0.016** - weak cross-architecture agreement on feature ordering.

rank	headline (tabpfn)	runner-up (xgboost)
1	headache_free_streak	days_since_last_migraine
2	days_since_last_migraine	headache_free_streak
3	migraine_rate_last7	migraine_rate_last7
4	menstruation_today	consecutive_sedentary_days
5	inappropriate_lighting	moderate_exercise_min
6	inappropriate_lighting_today	dow
7	migraine_rate_last3	sleep_variability_7day
8	migraine_yesterday	days_since_last_record
9	cheese_chocolate_today	exercise_days_7day
10	sleep_variability_7day	migraine_rate_last3

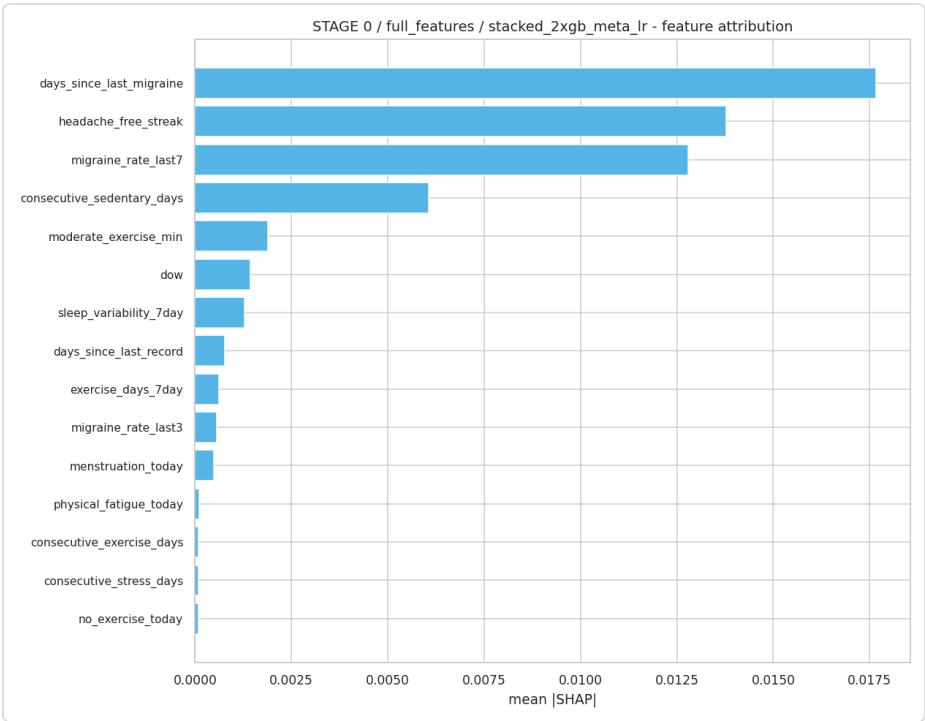
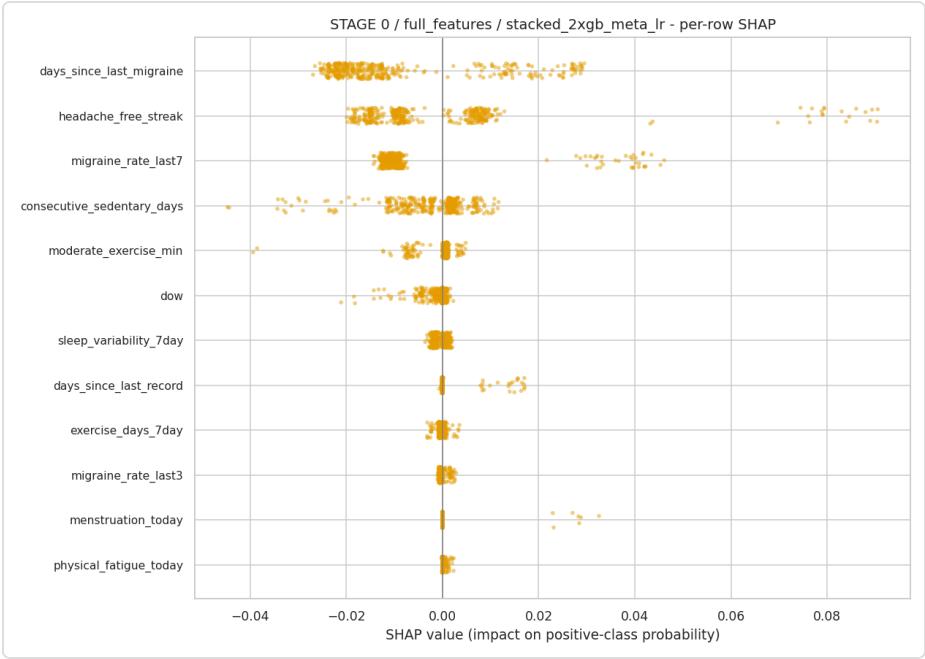
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_15_15/patient)



runner-up: xgboost stacked_2xgb_meta_lr (70_15_15/patient)



migraine / no_rolling_features

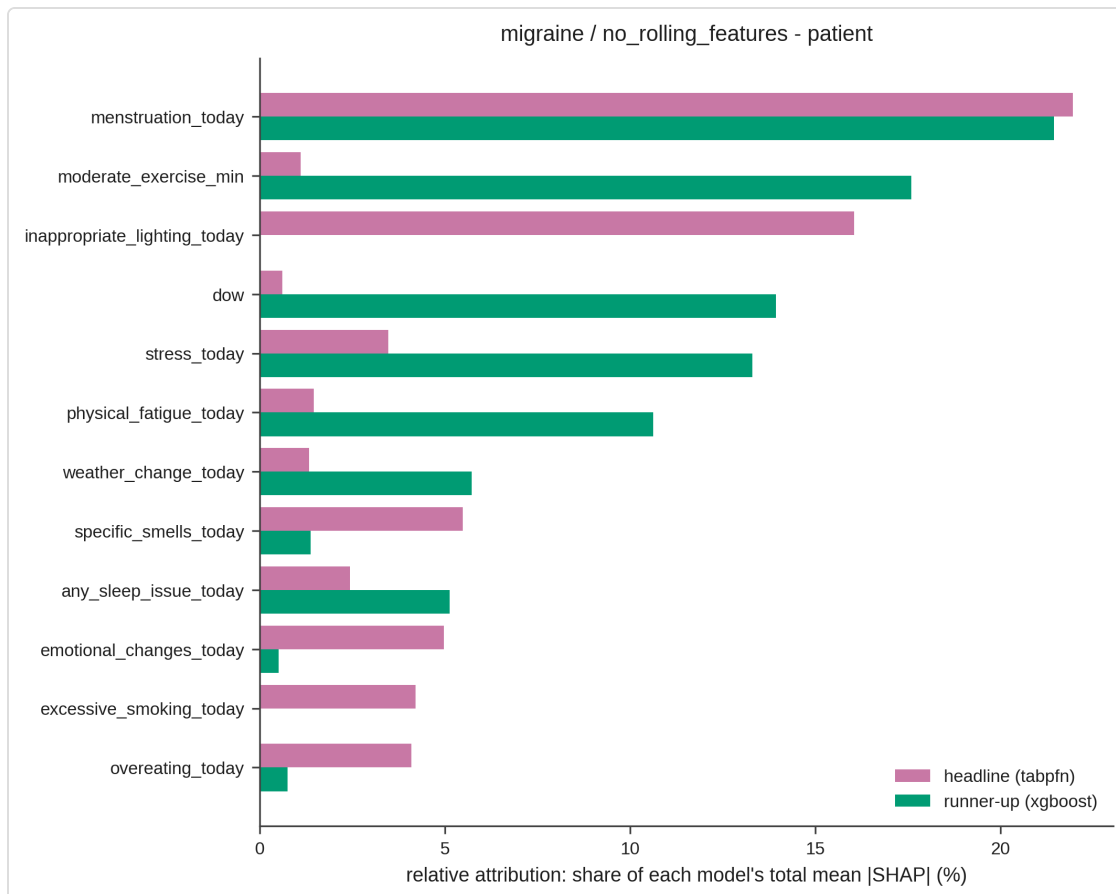
headline: **tabpfn/version_3-binary** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.599, AUPRC 0.103, calib +0.42
runner-up: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.496, AUPRC 0.082, calib -0.08

Spearman rank correlation over all 26 shared features: **-0.191** - weak cross-architecture disagreement on feature ordering.

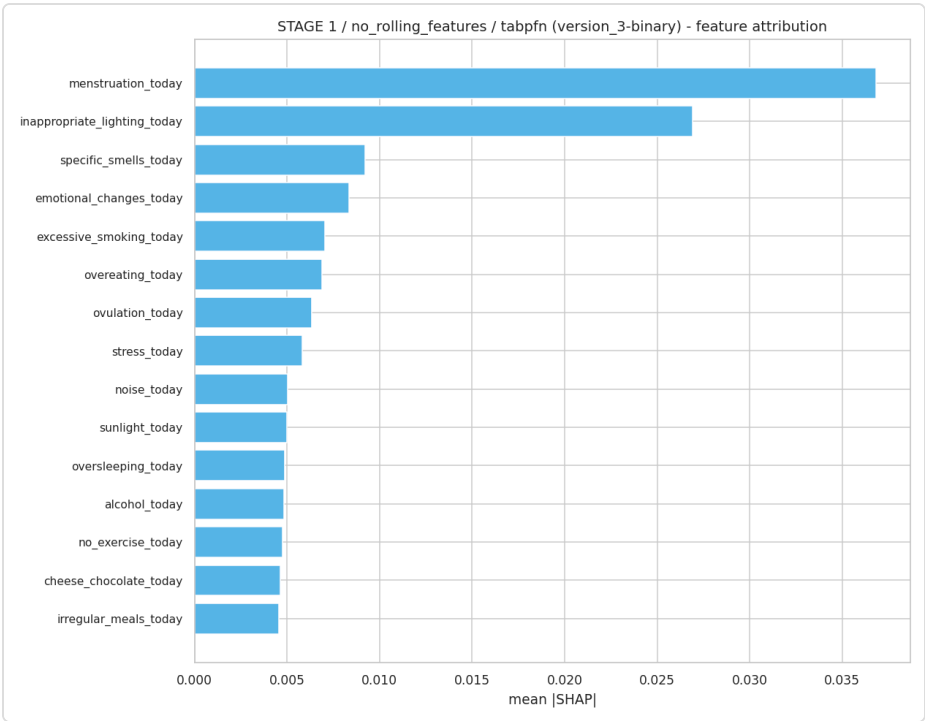
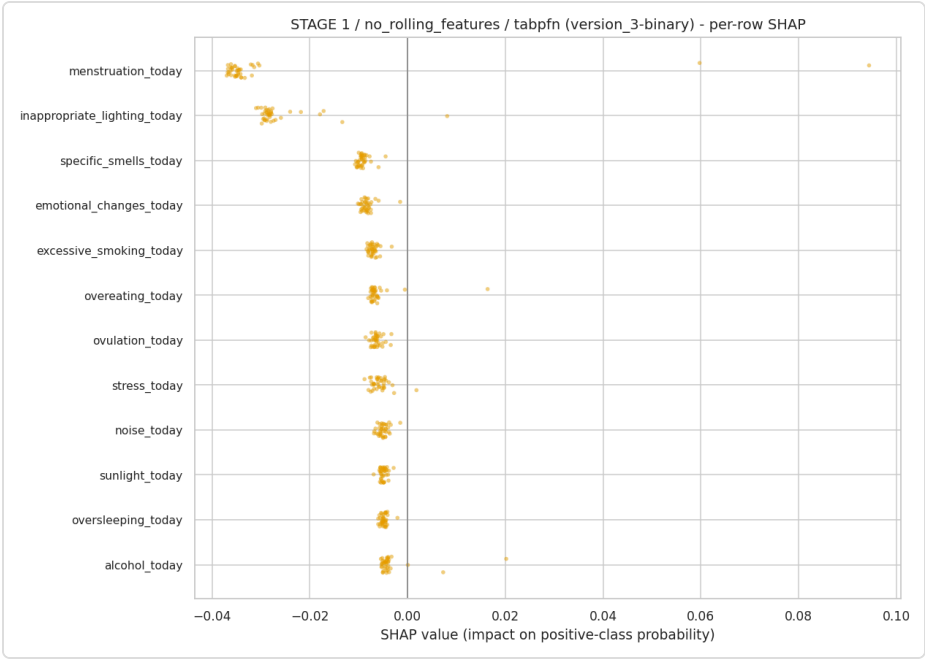
rank	headline (tabpfn)	runner-up (xgboost)
1	menstruation_today	menstruation_today

2	inappropriate_lighting_today	moderate_exercise_min
3	specific_smells_today	dow
4	emotional_changes_today	stress_today
5	excessive_smoking_today	physical_fatigue_today
6	overeating_today	weather_change_today
7	ovulation_today	any_sleep_issue_today
8	stress_today	irregular_meals_today
9	noise_today	no_exercise_today
10	sunlight_today	exercise_today

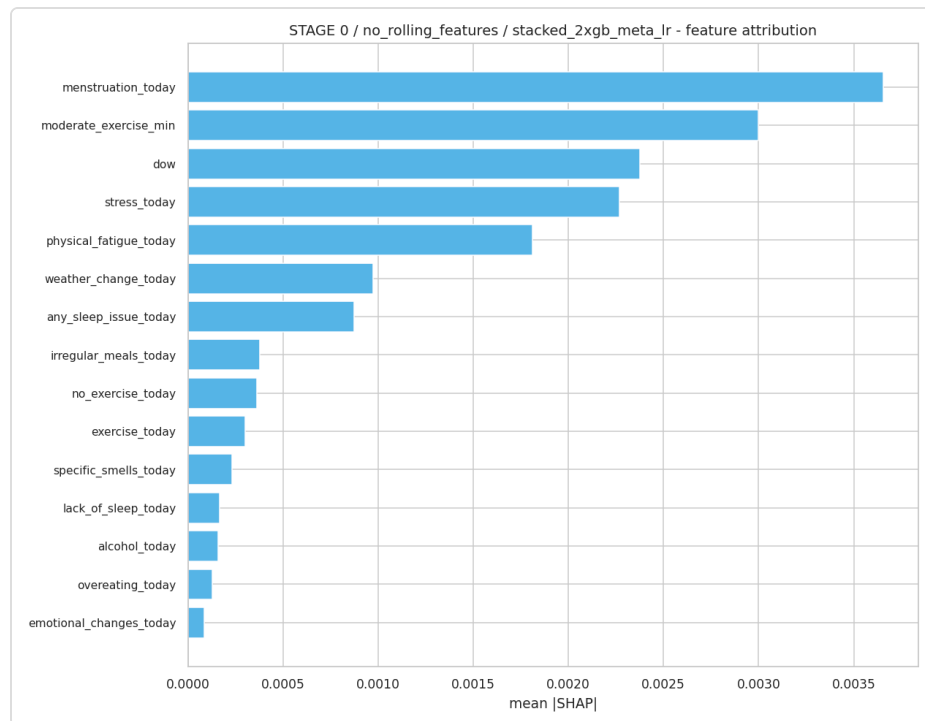
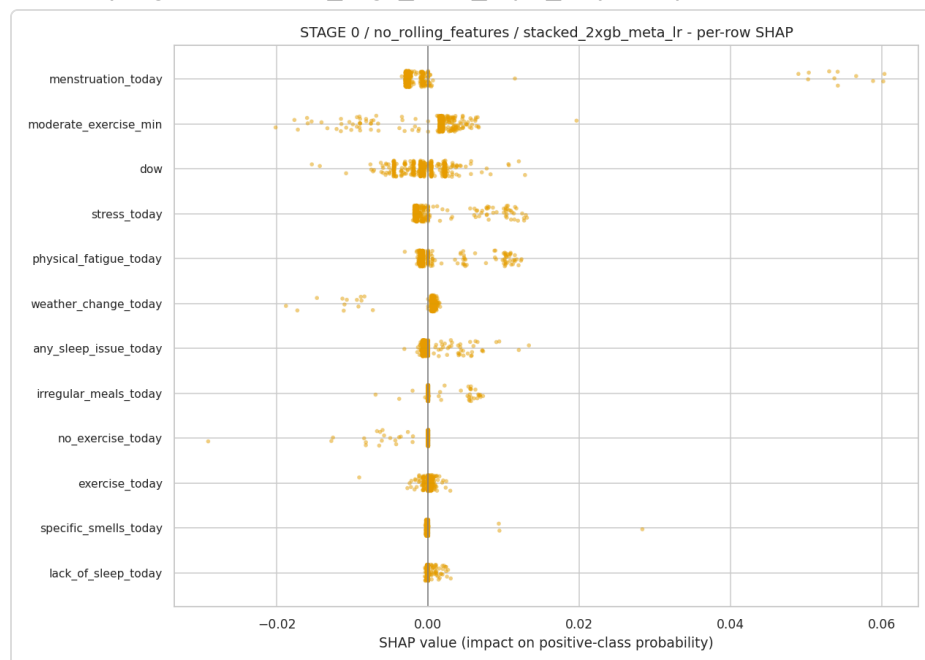
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: tabpfn tabpfn (70_30/patient)



runner-up: xgboost stacked_2xgb_meta_lr (70_30/patient)



migraine / park_features

This cell's headline is not significantly above chance (AUROC 95% CI [0.496-0.586] includes 0.5); its attributions describe a near-chance model and are not evidence of a real effect.

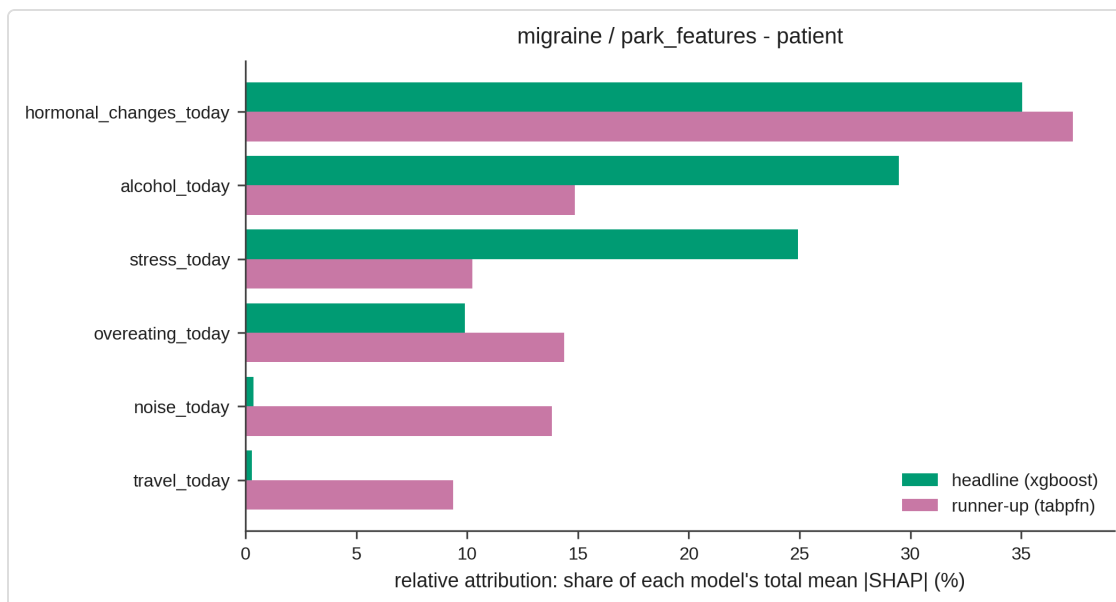
headline: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.541, AUPRC 0.086, calib +0.67

runner-up: **tabpfn/version_3-default** [NonHP (library defaults)] | split 70_30/patient | hold-out AUROC 0.536, AUPRC 0.082, calib +0.18

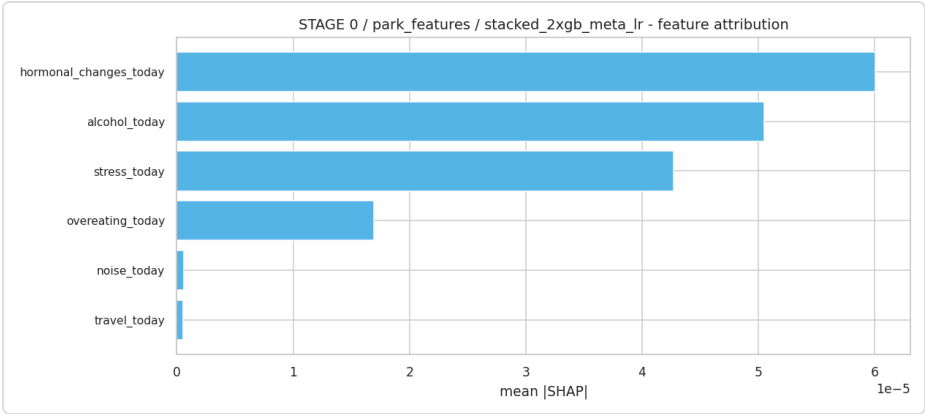
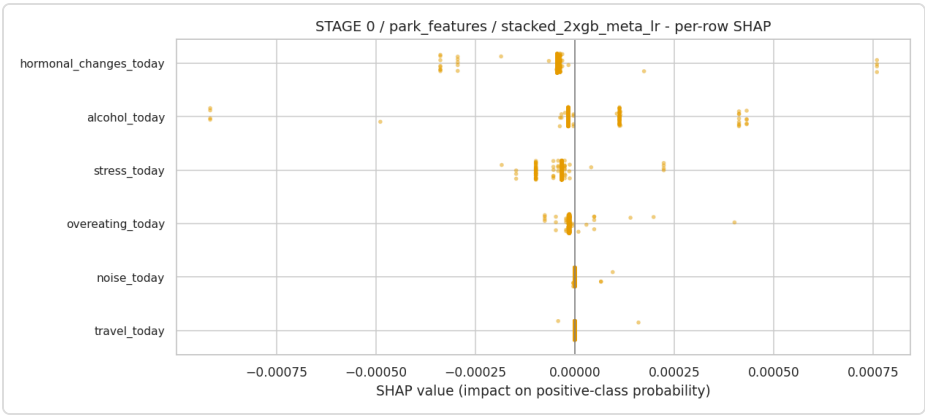
Spearman rank correlation over all 6 shared features: **+0.829** - strong cross-architecture agreement on feature ordering.

rank	headline (xgboost)	runner-up (tabpfn)
1	hormonal_changes_today	hormonal_changes_today
2	alcohol_today	alcohol_today
3	stress_today	overeating_today
4	overeating_today	noise_today
5	noise_today	stress_today
6	travel_today	travel_today
7		
8		
9		
10		

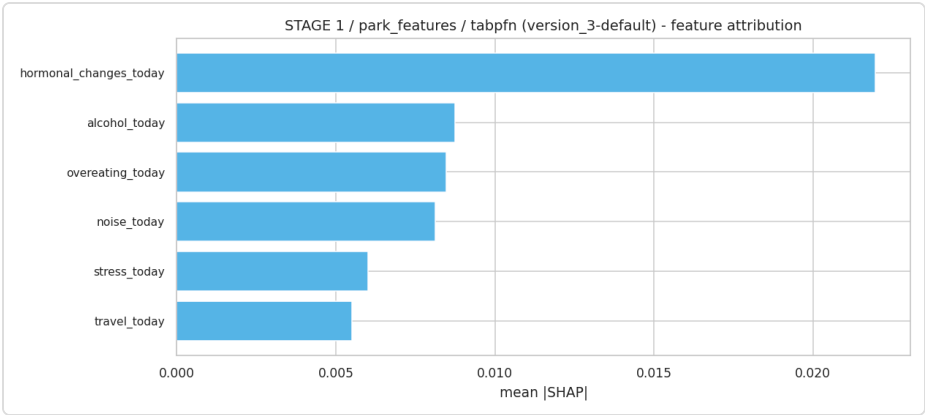
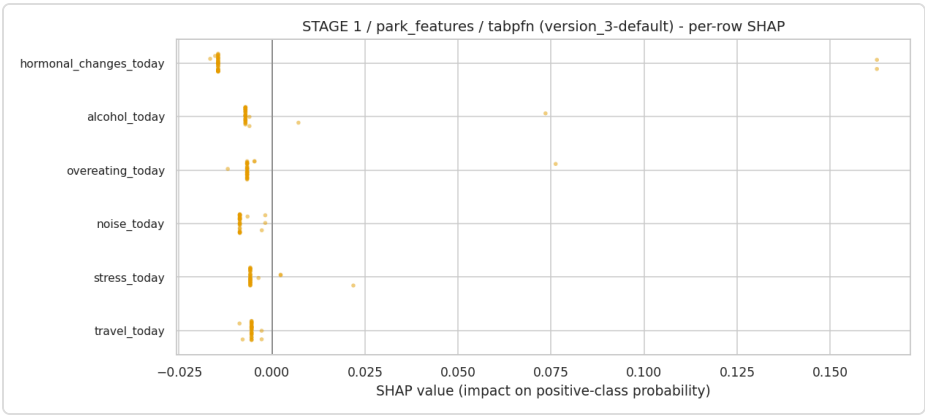
Cross-architecture attribution: each model's top features as a share of its own total mean |SHAP| (headline vs runner-up). Normalised because the two explainers' absolute magnitudes are not comparable; rank agreement is in the table above.



headline: xgboost stacked_2xgb_meta_lr (70_30/patient)



runner-up: tabpfn tabpfn (70_30/patient)



migraine / spano_features

Only one architecture insighted in this cell: **stacked_2xgb_meta_lr** [NonHP (library defaults)] | split 80_20/patient | hold-out AUROC 0.651, AUPRC 0.176, calib +4.96. No cross-family comparison available.

rank	feature (xgboost)	mean SHAP
1	days_since_last_migraine	0.0029
2	consecutive_sedentary_days	0.0013
3	sleep_variability_7day	0.0007
4	dow	0.0004
5	migraine_rate_last7	0.0004
6	consecutive_exercise_days	0.0001
7	menstruation_today	0.0001
8	stress_today	0.0001
9	exercise_today	0.0001
10	consecutive_stress_days	0.0001

leaf: xgboost stacked_2xgb_meta_lr (80_20/patient)

